All orders

Gyropup retained prototype workstreams · Revision: 0b7b5f0c0ef094898259c0b6f8a6ac52cd3a5feb

All parts: 279 of 279

Original leg, rig, turret and earlier body planning lists. Current front/rear-pack robot manufacturing plan remains incomplete. Checklist marks track parts, not purchased quantities. This list does not qualify fit, strength or a complete robot.

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/ballast_shaft	Tapped linear shaft 1 in x 6 in, 1050 steel, 3/8-16 x 3/4 in deep both ends	6649K433seller's no. 6649K433McMaster-Carr	qty2	each$52.94	line$105.88	1 alternate · 1 drop-inDROP-IN McMaster-Carr 4162N31 + 6435K18 collar · $38.42checked 2026-10-27	☑	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/circlip_50	External Retaining Ring, for 50 mm OD, Black-Phosphate Spring Steel	98541A167McMaster-Carr	qty1buy 1 pack of 5	each$3.10$15.50 per 5	line$15.50used $3.10	5 alternates · 5 drop-inDROP-IN Metric Specialties 50NCRE · $0.71 · available (Add to cart enabled; no count shown)checked 2026-10-27DROP-IN Aspen Fasteners ME304-50-00050 · Now: $183.21 (Box of 50) · Contact Support for Availability Infochecked 2026-10-27DROP-IN Fastenal 1135344 (package) / 35344 (bulk)checked 2026-10-27DROP-IN McMaster-Carr 90154A768 · $3.37 per pack of 1 · Delivers tomorrow 7-9 amZinc-chromate plated, sold singly: the	☐	☑

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							same DIN 471-50 ring and groove.checked 2026-10-27DROP-IN McMaster-Carr 90967A350 · $5.59 per pack of 1 · Delivers tomorrow 7-9 am1.4122 stainless, sold singly: the same DIN 471-50 ring and groove.checked 2026-10-27		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/circlip_60	External Retaining Ring, for 60 mm OD, Black-Phosphate Spring Steel	98541A174McMaster-CarrDelivers Wednesday 7-9 am (2026-10-27)	qty1	each$3.05	line$3.05	4 alternates · 4 drop-inDROP-IN Aspen Fasteners ME304-60-00040 · Now: $177.63 (Box of 40) · Contact Support for Availability Infochecked 2026-10-27DROP-IN Fastenal 35351checked 2026-10-27DROP-IN McMaster-Carr 90154A811 · $4.41 per pack of 1 · Delivers tomorrow 7-9 amZinc-chromate plated instead of black phosphate: the same DIN 471-60 ring and groove.checked 2026-10-27DROP-IN McMaster-Carr 90967A375 · $7.90 per pack of 1 · Delivers tomorrow 7-9 am1.4122 stainless	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							instead of spring steel: the same DIN 471-60 ring and groove; does not rust.checked 2026-10-27		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/dowel_5x16	Dowel Pin, 52100 Alloy Steel, 5 mm Diameter, 16 mm Long	91595A352McMaster-Carr	qty6buy 1 pack of 50	each$0.3844$19.22 per 50	line$19.22used $2.31	3 alternates · 3 drop-inDROP-IN WDS Components 662-20316 · $0.60 · Stock Guaranteed; stock availability < 500checked 2026-10-27DROP-IN Aspen Fasteners ME397-5X16-00300 · Now: $180.47 (Box of 300) · Contact Support for Availability Infochecked 2026-10-27DROP-IN Aspen Fasteners UB006-5X16-000400 · Now: $437.89 (Box of 400) · Contact Support for Availability Infochecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/dowel_5x20	Dowel Pin, 52100 Alloy Steel, 5 mm Diameter, 20 mm Long	91595A356McMaster-Carr	qty4buy 1 pack of 25	each$0.4352$10.88 per 25	line$10.88used $1.74	NO ALTERNATE	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/eye_sleeve	Metric Press-Fit Drill Bushing, 12 mm ID, 18 mm OD, 20 mm Long, Chamfer on One End	96990A403McMaster-CarrDelivers tomorrow 7-9 am (2026-10-27)	qty4	each$16.11	line$64.44	3 alternates · 1 drop-inDROP-IN JW Winco DIN 179-B12-20-A-NIchecked 2026-10-27CHANGES Reid Supply DM-12025 · $7.88 (1+) · In stock (9)25 mm long instead of 20 mm (same 12 mm ID and 18 mm OD): two end to end make 50 mm, so the knee eye must be 10 mm deeper, or one 25 mm with one 15 mm. ID and OD tolerance classes (F7 / n6) are not stated.checked 2026-10-27CHANGES McMaster-Carr 96990A240 · $37.07 each · Delivers tomorrow 7-9 am12 x 16 x 10 DIN 179 bushing: four end to end fill	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							the knee eye's 40 mm, in a Ø16 bore instead of Ø18, so the upper link's press bore changes from Ø17.96 to Ø15.96 (cad/fivebar.py EYE_SLEEVE). Four per eye at $37.07 each.checked 2026-10-27		
☐	buy/grommet_cable_pass	Buna-N Rubber Grommet, for 11/16" Hole Diameter and 1/4" Material Thicknes s, 7/16" ID	9307K884McMaster-Carr	qty1buy 1 pack of 10	each$1.08$10.82 per 10	line$10.82used $1.08	1 alternate · 1 drop-inDROP-IN printed product page · $0.00 · print a split TPU 95A grommet (as the deck's lidar pass)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/insert_m3	Heat-Set Threaded Inserts for Plastic, Standard, Brass, M3 x 0.5 mm Thread Size, 5.7 mm Installed Length	94459A140McMaster-Carr	qty97buy 2 packs of 50	each$0.2182$10.91 per 50	line$21.82used $21.17	6 alternates · 6 drop-inDROP-IN ruthex RX-M3x5.7 (SKU GE-M3x57-001) · InStock (schema.org availability)checked 2026-10-27DROP-IN 3DJake CNC-X001FCE4M5 · $11.18 ($0.11 / pc), excl. VAT · In stockchecked 2026-10-27DROP-IN 3DJake CNC-X001FCE4M5 · $11.18 / 100 · In stockchecked 2026-10-27DROP-IN ruthex RX-M3x5.7 (SKU GE-M3x57-001) · InStock (schema.org availability)checked 2026-10-27DROP-IN 3DJake CNC-X001FCE4M5 · $11.18 / 100 · In stockchecked 2026-10-	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							27DROP-IN ruthex RX-M3x5.7 (SKU GE-M3x57-001) · InStock (schema.org availability)checked 2026-10-27		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/insert_m5	Heat-Set Threaded Inserts for Plastic, Standard, Brass, M5 x 0.8 mm Thread Size, 7.9 mm Installed Length	94459A775seller's no. 94459A775McMaster-Carr	qty49buy 2 packs of 25	each$0.2728$6.82 per 25	line$13.64used $13.37	8 alternates · 6 drop-inCHANGES ruthex RX-M5x9.5 (SKU GE-M5x95-001) · InStock (schema.org availability)NOT drop-in: 9.5 mm long vs McMaster 7.9 mm (1.6 mm longer). The short version RX-M5Sx5.8 (and CNC Kitchen 'M5 x 5.8 short', also listed) is 2.1 mm shorter. No 7.9 mm M5 at ruthex or CNC Kitchen. EUR 9.99 per 50 (VAT incl.), no USD price.checked 2026-10-27CHANGES 3DJake CNC-B09FXM8ZMV · $11.18 ($0.22 / pc), excl. VAT · In stock5.8 mm long instead of 7.9 mm, so 2.1 mm less thread engagement	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							and pull-out strength. Same 6.4 mm hole per CNC Kitchen's table. It fits the existing 6.4 mm hole without a CAD change and sits flush with spare depth below.checked 2026-10-27DROP-IN ruthex RX-M5x9.5 (SKU GE-M5x95-001) · InStock (schema.org availability)checked 2026-10-27DROP-IN 3DJake CNC-B09FXM8ZMV · $11.18 / 50 · In stockchecked 2026-10-27DROP-IN 3DJake CNC-B09FXM8ZMV · $11.18 / 50 · In stockchecked 2026-10-27DROP-IN ruthex RX-M5x9.5 (SKU GE-M5x95-001) · InStock		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							(schema.org availability)checked 2026-10-27DROP-IN 3DJake CNC-B09FXM8ZMV · $11.18 / 50 · In stockchecked 2026-10-27DROP-IN ruthex RX-M5x9.5 (SKU GE-M5x95-001) · InStock (schema.org availability)checked 2026-10-27		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/knee_pin	Alloy Steel Shoulder Screws, 12 mm Shoulder Diameter, 90 mm Shoulder Length, M10 x 1.5 mm Thread	92981A411McMaster-CarrDelivers tomorrow 7-9 am (2026-10-27)	qty2	each$8.02	line$16.04	1 alternate · 1 drop-inDROP-IN Aspen Fasteners ME343-1015X90X12-00010 · $168.80 (box of 10) · Contact Support for Availability InfoISO 7379 12 x 90, M10 x 1.5, class 12.9 but plain (no black oxide): it rusts sooner. Sold only by the box of 10.checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/knee_spacer	Multipurpose 6061 Aluminum Round Tube, 0.065" Wall Thicknes s, 5/8" OD	9056K68McMaster-Carr	qty2buy 1 length, cut 4 per length	each$6.26 / length$1.57 a piece	line$6.26used $3.13	3 alternates · 3 drop-inDROP-IN Speedy Metals t61r.625x.065 · 12in length: $8.96 (custom by the inch: Base Price $0.50)checked 2026-10-27DROP-IN Alcobra Metals 22210129bf34 · $20.03 (12in) · In stockchecked 2026-10-27DROP-IN McMaster-Carr 90018A130 · $70.39 each · Delivers in 1-3 weeksMade to order, no lathe needed: order it 0.630 in OD x 0.500 in ID x 1.102 in (28.0 mm) long (16.0 x 12.7 mm, under the 17 mm limit, clears the 12 mm shoulder). About $70 each at 1-4 against $1.56 of tube, and 1-3	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							weeks.checked 2026-10-27		
☐	buy/nut_m10_nyloc	High-Strength Steel Nylon-Insert Locknut, Zinc Plated, M10 x 1.5 mm Thread Size	94645A220McMaster-CarrDelivers Wednesday 7-9 am (2026-10-27)	qty2buy 1 pack of 25	each$0.4632$11.58 per 25	line$11.58used $0.93	2 alternates · 2 drop-inDROP-IN Belmetric NN10CLZ · $0.26 · In stockchecked 2026-10-27DROP-IN Aspen Fasteners ME022-1015 · Now: $148.51 · Contact Support for Availability Infochecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/nut_m3	Steel Hex Nut, Medium-Strength, Class 8, M3 x 0.5 mm (held in the P3 shell's nut pads)	90592A085seller's no. 90592A085McMaster-Carrproduct page read (McMaster ships next day) (2026-10-27)	qty10buy 1 pack of 100	each$0.0294$2.94 per 100	line$2.94used $0.29	1 alternate · 1 drop-inDROP-IN McMaster-Carr 91828A211 · $5.68 / 100 · 18-8 stainless M3 hex nut, the same sizechecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/nut_m5_nyloc	High-Strength Steel Nylon-Insert Locknut, Zinc Plated, M5 x 0.8 mm Thread Size	94645A102McMaster-Carr	qty10buy 1 pack of 100	each$0.1566$15.66 per 100	line$15.66used $1.57	2 alternates · 2 drop-inDROP-IN Belmetric NN5CLZ · $0.09 · in stockchecked 2026-10-27DROP-IN Belmetric NN5CLZ · $0.09 · in stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/nut_m6_nyloc	High-Strength Steel Nylon-Insert Locknut, Zinc Plated, M6 x 1 mm Thread Size	94645A205McMaster-Carr	qty10buy 1 pack of 100	each$0.1226$12.26 per 100	line$12.26used $1.23	1 alternate · 1 drop-inDROP-IN Global Industrial WBB2298457 (Titan Fasteners AAV06) · $25.95 / 100 · ships in 2 business days: DIN 985 M6 in 304 stainless (the same size, 6 mm tall)checked 2026-10-27	☐	☐
☐	buy/pin_lanyard	Lanyard, 3/64 in cable, loop-to-eye, 12 in	90312A296seller's no. 90312A296McMaster-Carr	qty1	each$3.26	line$3.26	1 alternate · 1 drop-inDROP-IN Essentra Components LN3/64120003 BGP · $2.90 · 900 In stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/quick_pin	Ball-lock quick-release pin with D-ring, 3/8 in, 1-1/2 in usable, 17-4 PH	92384A274seller's no. 92384A274McMaster-Carr	qty1	each$38.21	line$38.21	1 alternate · 1 drop-inDROP-IN McMaster-Carr 92384A760 (1-3/4 in usable) · $40.73checked 2026-10-27	☐	☐
☐	buy/screw_3_8x1	Alloy steel socket head screw 3/8-16 x 1 in	91251A624seller's no. 91251A624McMaster-Carr	qty2buy 1 pack of 25	each$0.4992$12.48 per 25	line$12.48used $1.00	1 alternate · 1 drop-inDROP-IN McMaster-Carr 91251A622 (3/4 in, no washer) · $11.17 / 25checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/screw_3_8x3_4	Alloy steel socket head screw 3/8-16 x 3/4 in	91251A622seller's no. 91251A622McMaster-Carr	qty2buy 1 pack of 25	each$0.4468$11.17 per 25	line$11.17used $0.89	1 alternate · 1 drop-inDROP-IN Bolt Depot 5044 · $0.34 · no count shown; orderable (Add button, not marked 'Temporarily unavailable')checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/screw_m3x16	Black-Oxide Alloy Steel Socket Head Screw, M3 x 0.5 x 16 mm (the P3 shell: walls and faces into the posts' inserts, the door, hatch and	91290A120seller's no. 91290A120McMaster-Carrproduct page read (McMaster ships next day) (2026-10-27)	qty16buy 1 pack of 100	each$0.1222$12.22 per 100	line$12.22used $1.96	1 alternate · 1 drop-inDROP-IN Belmetric BSH3X16 · $0.16 · in stock: DIN 912 12.9 black, the same screw; $0.10 each from 25checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
		splices)							
☐	buy/screw_m3x6	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M3 x 0.5 mm Thread Size, 6 mm Long	91290A111McMaster-Carr	qty45buy 1 pack of 100	each$0.1159$11.59 per 100	line$11.59used $5.22	3 alternates · 3 drop-inDROP-IN Belmetric BSH3X6 · $0.12 · in stockchecked 2026-10-27DROP-IN Belmetric BSH3X6 · $0.12 · in stockchecked 2026-10-27DROP-IN Belmetric BSH3X6 · $0.12 · in stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/screw_m3x8_socket	Black-oxide alloy steel socket head screw, M3 x 0.5 x 8 mm (the bus bars to their insulators, the insulators to the plate)	91290A113seller's no. 91290A113McMaster-Carrproduct page read (McMaster ships next day) (2026-10-27)	qty26buy 1 pack of 100	each$0.1328$13.28 per 100	line$13.28used $3.45	2 alternates · 2 drop-inDROP-IN McMaster-Carr 91290A115 · $13.23 / 100 · M3 x 10 of the same serieschecked 2026-10-27DROP-IN McMaster-Carr 91290A115 · $13.23 / 100 · M3 x 10 of the same serieschecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/screw_m4x10_flat	Black-oxide alloy steel flat head hex drive screw, 90 deg, M4 x 0.7 x 10 mm (the spine standoffs, flush on the frame side)	91294A190seller's no. 91294A190McMaster-Carrproduct page read (McMaster ships next day) (2026-10-27)	qty16buy 1 pack of 100	each$0.0638$6.38 per 100	line$6.38used $1.02	1 alternate · 1 drop-inDROP-IN McMaster-Carr 91294A188 · $6.25 / 100 · M4 x 8 of the same series (engages 3.3 mm in the standoff)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/screw_m4x12_socket	Black-oxide alloy steel socket head screw, M4 x 0.7 x 12 mm	91290A148seller's no. 91290A148McMaster-Carrproduct page read (McMaster ships next day) (2026-10-27)	qty14buy 1 pack of 100	each$0.123$12.30 per 100	line$12.30used $1.72	2 alternates · 2 drop-inDROP-IN McMaster-Carr 91290A150 · $14.23 / 100 · M4 x 14 of the same series (a stack of three lugs)checked 2026-10-27DROP-IN McMaster-Carr 91290A150 · $14.23 / 100 · M4 x 14 of the same series (a stack of three lugs)checked 2026-10-27	☐	☐
☐	buy/screw_m4x16_rail	18-8 stainless socket head screw M4 x 16	91292A118seller's no. 91292A118McMaster-Carr	qty36buy 1 pack of 100	each$0.1163$11.63 per 100	line$11.63used $4.19	1 alternate · 1 drop-inDROP-IN Bolt Depot 23082 (class 12.9, zinc) · $0.19checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/screw_m4x20	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M4 x 0.7 mm Thread Size, 20 mm Long	91290A168McMaster-Carr	qty32buy 1 pack of 100	each$0.1447$14.47 per 100	line$14.47used $4.63	2 alternates · 2 drop-inDROP-IN Belmetric BSH4X20 · $0.20 · In stockchecked 2026-10-27DROP-IN Belmetric BSH4X20 · $0.20 · In stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/screw_m5x10_button	Black-oxide class 10.9 button head hex drive screw, ISO 7380, M5 x 0.8 x 10 mm (tray, belly, posts, VESCs, bay)	91239A224seller's no. 91239A224McMaster-Carrproduct page read (McMaster ships next day) (2026-10-27)	qty18buy 1 pack of 100	each$0.1179$11.79 per 100	line$11.79used $2.12	2 alternates · 2 drop-inDROP-IN McMaster-Carr 91239A228 · $12.06 / 100 · M5 x 12 of the same serieschecked 2026-10-27DROP-IN McMaster-Carr 91239A228 · $12.06 / 100 · M5 x 12 of the same serieschecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/screw_m5x12_button	Black-oxide class 10.9 button head hex drive screw, ISO 7380, M5 x 0.8 x 12 mm (the hip frames into the cheeks)	91239A228seller's no. 91239A228McMaster-Carrproduct page read (McMaster ships next day) (2026-10-27)	qty40buy 1 pack of 100	each$0.1206$12.06 per 100	line$12.06used $4.82	1 alternate · 1 drop-inDROP-IN McMaster-Carr 91239A224 · $11.79 / 100 · M5 x 10 of the same series (engages 5.3 mm: only if the 12 is out)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/screw_m5x14_socket	Black-oxide alloy steel socket head screw, M5 x 0.8 x 14 mm (the fuse board's plate bond, MH1, into the plate's flush nut)	91290A230seller's no. 91290A230McMaster-Carrproduct page read (McMaster ships next day) (2026-10-27)	qty8buy 1 pack of 100	each$0.1743$17.43 per 100	line$17.43used $1.39	1 alternate · 1 drop-inDROP-IN McMaster-Carr 91290A228 · $16.36 / 100 · M5 x 12 of the same series, with MH1's standoff printed 1.6 mm shorterchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/screw_m5x16	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M5 x 0.8 mm Thread Size, 16 mm Long	91290A232seller's no. 91290A232McMaster-Carr	qty10buy 1 pack of 100	each$0.1727$17.27 per 100	line$17.27used $1.73	2 alternates · 2 drop-inDROP-IN Belmetric BSH5X16 · $0.17 · in stockchecked 2026-10-27DROP-IN Bolt Depot 6522 · $0.14 · no count shown; orderable (Add button, not marked 'Temporarily unavailable')checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/screw_m5x25	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M5 x 0.8 mm Thread Size, 25 mm Long	91290A252seller's no. 91290A252McMaster-Carr	qty36buy 1 pack of 50	each$0.2192$10.96 per 50	line$10.96used $7.89	1 alternate · 1 drop-inDROP-IN Bolt Depot 6524 (M5 x 25 class 12.9 black oxide) · $0.15checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/screw_m5x40	Black-Oxide Alloy Steel Socket Head Screw, M5 x 0.8 mm Thread Size, 40 mm Long, Partially Threaded	91290A662seller's no. 91290A662McMaster-Carr	qty19buy 1 pack of 50	each$0.1618$8.09 per 50	line$8.09used $3.07	4 alternates · 4 drop-inDROP-IN Belmetric BSH5X40 · $0.28 · In stockchecked 2026-10-27DROP-IN McMaster-Carr 91290A258 · $16.58 per pack of 50 · Delivers tomorrow 7-9 amThe US-origin column of the same class 12.9 ISO 4762 table: same screw, about twice the price.checked 2026-10-27DROP-IN Belmetric BSH5X40 · $0.28 · In stockchecked 2026-10-27DROP-IN McMaster-Carr 91290A258 · $16.58 / 50 · Delivers tomorrow 7-9 amchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/screw_m6x16	Black-Oxide Alloy Steel Socket Head Screw, M6 x 1 mm Thread Size, 16 mm Long, Fully Threaded	91290A667McMaster-Carr	qty10buy 1 pack of 100	each$0.0859$8.59 per 100	line$8.59used $0.86	1 alternate · 1 drop-inDROP-IN Belmetric BSH6X16 · $0.20 · in stock: DIN 912 class 12.9, black oxide, fully threaded (the same screw); $0.12 each from 25checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/screw_m6x30	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M6 x 1 mm Thread Size, 30 mm Long	91290A332seller's no. 91290A332McMaster-Carr	qty12buy 1 pack of 50	each$0.2272$11.36 per 50	line$11.36used $2.73	2 alternates · 2 drop-inDROP-IN Belmetric BSH6X30 · $0.23 · In stockchecked 2026-10-27DROP-IN Belmetric BSH6X30 · $0.23 · In stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/screw_m6x35	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M6 x 1 mm Thread Size, 35 mm Long	91290A334seller's no. 91290A334McMaster-Carr	qty4buy 1 pack of 50	each$0.2422$12.11 per 50	line$12.11used $0.97	1 alternate · 1 drop-inDROP-IN Bolt Depot 6543 · $0.24 · no count shown; orderable (Add button, not marked 'Temporarily unavailable')checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/shoulder_6x12_m5	Alloy Steel Shoulder Screws, 6 mm Shoulder Diameter, 12 mm Shoulder Length, M5 x 0.8 mm Thread	92981A101McMaster-Carr	qty1	each$2.82	line$2.82	2 alternates · 1 drop-inDROP-IN Aspen Fasteners ME343-508X12X6-00080 · Now: $191.06 (Box of 80) · Contact Support for Availability Infochecked 2026-10-27CHANGES Belmetric BSHLDR5X6X10/19 · $4.39 · In stockShoulder length is 10 mm instead of 12 mm (2 mm shorter). The clamped stack must be 2 mm thinner or the CAD changed. The 6 mm shoulder, M5 x 0.8 thread and class 12.9 black oxide all match.checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/standoff_m3x16_ff	Lightweight aluminium female-threaded hex standoff, 6 mm hex, M3 x 16 mm (the J501 carrier on the pallet)	95947A048seller's no. 95947A048McMaster-Carrproduct page read (McMaster ships next day) (2026-10-27)	qty4	each$0.810	line$3.24	1 alternate · 1 drop-inDROP-IN McMaster-Carr 95947A051 · $0.81 · M3 x 19 of the same series (the carrier 3 mm higher)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/standoff_m4x31_ff	Lightweight aluminium female-threaded hex standoff, 6 mm hex, M4 x 31 mm (the spine: between the cheeks)	95947A065seller's no. 95947A065McMaster-Carrproduct page read (McMaster ships next day) (2026-10-27)	qty8	each$1.46	line$11.68	1 alternate · 1 drop-inDROP-IN McMaster-Carr 95947A062 · $1.35 · M4 x 30 of the same series, with a printed 1 mm washer at one endchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/stop_pin	Vented Pull-Out Dowel Pin with Flat Shaft, High-Strength 52100 Alloy Steel, 16 mm Diameter, 40 mm Long	98432A386McMaster-CarrDelivers Thursday (2026-10-27)	qty4	each$5.88	line$23.52	2 alternates · none drop-inCHANGES McMaster-Carr 91595A845 · $5.01 eachThe plain 52100 dowel (DIN 6325-m6, Rockwell C52, Ø16 x 40): no vent and no tapped end. Print a Ø2 vent through the hole's 1.1 mm floor at its outer edge (off the motor's face) so air and compound get out, and accept that the pin cannot be pulled out of the blind hole.checked 2026-10-27CHANGES Aspen Fasteners ME397-16X40-00008 · $192.49 (box of 8) · Contact Support for Availability InfoA plain DIN 6325 dowel	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							(m6, hardened Rockwell C58-C62, rounded ends): the same changes as the plain McMaster pin; four times the price.checked 2026-10-27		
☐	buy/tool-circlip-pliers	External retaining-ring pliers for a 50 mm ring (tool)	5449A83seller's no. 5449A83McMaster-Carr'Delivers tomorrow 7-9 am' (McMaster's in-stock wording) (2026-10-27)	qty1	each$37.25	line$37.25	1 alternate · 1 drop-inDROP-IN McMaster-Carr 5449A82 · $33.48 · listed in the category table (not opened; McMaster stocks the family)checked 2026-10-27	☐	☐
☐	buy/tool-dial-indicator	Dial indicator with a jointed-arm magnetic base (tool)	5518N11seller's no. 5518N11McMaster-Carr'Delivers tomorrow 7-9 am' (2026-10-27)	qty1	each$79.62	line$79.62	1 alternate · 1 drop-inDROP-IN McMaster-Carr 6302N12 · $84.93 · 'Delivers tomorrow 7-9 am'checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/tool-insert-tips	Heat-set insert installation tip set, M2-M8, for a Hakko 900 M-style iron (tool)	8915N111seller's no. 8915N111McMaster-Carr'Delivers tomorrow 7-9 am' (2026-10-27)	qty1	each$22.66	line$22.66	2 alternates · 2 drop-inDROP-IN McMaster-Carr 97338A140 · $74.43 · 'Delivers tomorrow 7-9 am'checked 2026-10-27DROP-IN McMaster-Carr 92160A315 + 92160A323 + 92160A327 · $28.14 / 3 · listed in the category table (product pages not opened)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/tool-reamer-5mm	Chucking reamer, 5 mm, turned by hand in a tap wrench (tool)	8851A18seller's no. 8851A18McMaster-Carrlisted in the category table (product page not opened) (2026-10-27)	qty1	each$26.23	line$26.23	1 alternate · 1 drop-inDROP-IN McMaster-Carr 2822A37 · $106.53 · 'Delivers tomorrow 7-9 am'checked 2026-10-27	☐	☐
☐	buy/turret_dowel_4x8	Dowel Pin, 52100 Alloy Steel, 4 mm Diameter, 8 mm Long	91595A150McMaster-Carr	qty2buy 1 pack of 50	each$0.242$12.10 per 50	line$12.10used $0.48	1 alternate · 1 drop-inDROP-IN WDS Components 662-20208 · $0.49 · stock availability < 500checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/turret_insert_m3_short	Heat-Set Threaded Inserts for Plastic, Standard, Brass, M3 x 0.5 mm Thread Size, 4.3 mm Installed Length	94459A130McMaster-Carr	qty2buy 1 pack of 50	each$0.2182$10.91 per 50	line$10.91used $0.44	1 alternate · 1 drop-inDROP-IN ruthex RX-M3Sx4.0 (SKU GE-M3Sx40-002) · InStock (schema.org availability)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/turret_insert_m4	Heat-Set Threaded Inserts for Plastic, Standard, Brass, M4 x 0.7 mm Thread Size, 6.4 mm Installed Length	94459A160seller's no. 94459A160McMaster-Carr	qty16buy 1 pack of 50	each$0.2346$11.73 per 50	line$11.73used $3.75	2 alternates · 1 drop-inCHANGES ruthex RX-M4x8.1 (SKU GE-M4x81-001) · InStock (schema.org availability)NOT drop-in: 8.1 mm long vs McMaster 6.4 mm (1.7 mm longer, needs a deeper boss). The other ruthex option, RX-M4Sx4.0 short, is 2.4 mm shorter (page not opened). Neither ruthex nor CNC Kitchen (M4x8.1 / M4x4) sells a 6.4 mm M4. EUR 9.49 per 50 (VAT incl.), no USD price.checked 2026-10-27DROP-IN Fastenal 11125996 (E-Z LOK / E-Z Sonic TH-M40-SV) · $0.24 · Local	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							DC Inventory; Direct Ship: Ships TOMORROWchecked 2026-10-27		
☐	buy/turret_magnet_disc	Neodymium Magnet, Magnetized Through Thickness, 1/8" Thick, 3/16" OD	5862K102McMaster-Carr	qty1	each$1.00	line$1.00	1 alternate · 1 drop-inDROP-IN K&J Magnetics D32-N52 · $0.35 · In Stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/turret_nut_m2_5	Corrosion-Resistant 18-8 Stainless Steel Hex Nut, M2.5 x 0.45 mm Thread Size	91828A113McMaster-Carr	qty2buy 1 pack of 100	each$0.071$7.10 per 100	line$7.10used $0.14	1 alternate · 1 drop-inDROP-IN Belmetric NR2.5SS · $0.06 · in stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/turret_oring_ct	Oil-Resistant Buna-N O-Ring, 1/16 Fractional Width, Dash Number 006	9452K14McMaster-Carr	qty1buy 1 pack of 100	each$0.0279$2.79 per 100	line$2.79used $0.03	1 alternate · 1 drop-inDROP-IN The O-Ring Store LLC B70006 · $0.03 · 368780 in Stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/turret_screw_m2_5x8	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M2.5 x 0.45 mm Thread Size, 8 mm Long	91290A102seller's no. 91290A102McMaster-Carr	qty6buy 1 pack of 50	each$0.246$12.30 per 50	line$12.30used $1.48	2 alternates · 2 drop-inDROP-IN The Nutty Company MSHC2.58 · $0.12 · in stock (no count shown)checked 2026-10-27DROP-IN Bolt Depot 13330 · $0.20 · no count shown; orderable (Add button, not marked 'Temporarily unavailable')checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/turret_screw_m2x10	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M2 x 0.4 mm Thread Size, 10 mm Long	91290A017McMaster-Carr	qty4buy 1 pack of 100	each$0.1589$15.89 per 100	line$15.89used $0.64	1 alternate · 1 drop-inDROP-IN The Nutty Company MSHC210 · $0.13 · in stock (no count shown)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/turret_screw_m2x20	Black-Oxide Alloy Steel Socket Head Screw, M2 x 0.4 mm Thread Size, 20 mm Long	91290A049McMaster-Carr	qty4buy 1 pack of 100	each$0.1658$16.58 per 100	line$16.58used $0.66	1 alternate · 1 drop-inDROP-IN The Nutty Company MSHC220 · $0.18 · in stock (no count shown)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/turret_screw_m2x6_flat	18-8 Stainless Steel Hex-Drive Flat Head Screw, M2 x 0.4 mm Thread Size, 6 mm Long	92125A052McMaster-Carr	qty4buy 1 pack of 50	each$0.240$12.00 per 50	line$12.00used $0.96	1 alternate · 1 drop-inDROP-IN Metric Specialties 2C6MXF2 · $0.07 · availablechecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/turret_screw_m3x10	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M3 x 0.5 mm Thread Size, 10 mm Long	91290A115McMaster-Carr	qty10buy 1 pack of 100	each$0.1323$13.23 per 100	line$13.23used $1.32	2 alternates · 2 drop-inDROP-IN Belmetric BSH3X10 · $0.14 · in stockchecked 2026-10-27DROP-IN Belmetric BSH3X10 · $0.14 · in stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/turret_screw_m3x10_flat	Black-Oxide Alloy Steel Hex-Drive Flat Head Screw, 90 Degree Countersink, M3 x 0.5 mm Thread Size, 10 mm Long	91294A130McMaster-Carr	qty3buy 1 pack of 100	each$0.0636$6.36 per 100	line$6.36used $0.19	1 alternate · 1 drop-inDROP-IN Belmetric SF3X10 · $0.18 · in stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/turret_screw_m3x14_flat_ss	18-8 Stainless Steel Hex-Drive Flat Head Screw, M3 x 0.5 mm Thread Size, 14 mm Long	92125A133McMaster-Carr	qty4buy 1 pack of 100	each$0.1065$10.65 per 100	line$10.65used $0.43	1 alternate · 1 drop-inDROP-IN Belmetric SF3X14CLZ · $0.18 · in stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/turret_screw_m3x16	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M3 x 0.5 mm Thread Size, 16 mm Long	91290A120McMaster-Carr	qty4buy 1 pack of 100	each$0.1222$12.22 per 100	line$12.22used $0.49	1 alternate · 1 drop-inDROP-IN Belmetric BSH3X16 · $0.16 · in stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/turret_screw_m3x6_flat	Black-Oxide Alloy Steel Hex-Drive Flat Head Screw, 90 Degree Countersink, M3 x 0.5 mm Thread Size, 6 mm Long	91294A126McMaster-Carr	qty6buy 1 pack of 100	each$0.0545$5.45 per 100	line$5.45used $0.33	2 alternates · 2 drop-inDROP-IN Belmetric SF3X6 · $0.18 · in stockchecked 2026-10-27DROP-IN Belmetric SF3X6 · $0.18 · in stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/turret_screw_m3x8	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M3 x 0.5 mm Thread Size, 8 mm Long	91290A113McMaster-Carr	qty23buy 1 pack of 100	each$0.1328$13.28 per 100	line$13.28used $3.05	1 alternate · 1 drop-inDROP-IN Belmetric BSH3X8 · $0.13 · in stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/turret_screw_m3x8_button	Class 10.9 Screw, M3 x 0.5 mm Thread Size, 8 mm Long	91239A113McMaster-Carr	qty8buy 1 pack of 100	each$0.0982$9.82 per 100	line$9.82used $0.79	1 alternate · 1 drop-inDROP-IN Fastener Superstore 227258 · $31.01 / carton of 100 · not shownchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/turret_screw_m3x8_flat	Black-Oxide Alloy Steel Hex-Drive Flat Head Screw, 90 Degree Countersink, M3 x 0.5 mm Thread Size, 8 mm Long	91294A128McMaster-Carr	qty4buy 1 pack of 100	each$0.0582$5.82 per 100	line$5.82used $0.23	1 alternate · 1 drop-inDROP-IN Belmetric SF3X8 · $0.18 · in stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/turret_screw_m4x10	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M4 x 0.7 mm Thread Size, 10 mm Long	91290A144seller's no. 91290A144McMaster-Carr	qty24buy 1 pack of 100	each$0.1355$13.55 per 100	line$13.55used $3.25	2 alternates · 2 drop-inDROP-IN Belmetric BSH4X10 · $0.14 · in stockchecked 2026-10-27DROP-IN Bolt Depot 6506 · $0.12 · no count shown; orderable (Add button, not marked 'Temporarily unavailable')checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/turret_set screw_m3x6_nylon	Nylon-Tip 18-8 Stainless Steel Set Screw, M3 x 0.5 mm Thread, 6 mm Long	93285A115McMaster-Carr	qty4buy 1 pack of 5	each$3.46$17.31 per 5	line$17.31used $13.85	1 alternate · 1 drop-inDROP-IN Belmetric SS3X6NYTPSS · $3.25 · in stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/turret_standoff_m3x12	Aluminum Male-Female Threaded Hex Standoff, 4.5 mm Hex, 12mm Long, M3 x 0.50 mm Thread	98952A110McMaster-Carr	qty4	each$0.830	line$3.32	1 alternate · 1 drop-inDROP-IN Mouser 761-M2107-3005-AL · $0.53 · 691checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/turret_standoff_m3x14	Aluminum Male-Female Threaded Hex Standoff, 4.5 mm Hex, 14mm Long, M3 x 0.50 mm Thread	98952A112McMaster-Carr	qty4	each$0.840	line$3.36	1 alternate · 1 drop-inDROP-IN Mouser 761-M2109-3005-AL · $1.42 · 73checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/turret_thumbscrew_m4x12	Steel Raised Knurled-Head Thumb Screw, M4 x 0.7 mm Thread Size, 12mm Long	92581A250McMaster-Carr	qty1	each$4.64	line$4.64	1 alternate · 1 drop-inDROP-IN Belmetric BTHUMB4X12SS · $2.46 · in stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/turret_tube_coupling	Continuous-Flex Tpv Rubber Tube for Chemicals, 3/16" ID, 5/16" OD	51225K24McMaster-Carr	qty1	each$0.620	line$0.62	1 alternate · 1 drop-inDROP-IN Zoro Zoro Select ZUSA-HT-3487 (Zoro # G8076328) · $12.49checked 2026-10-27	☐	☐
☐	buy/washer_2in	Oversized washer 0.406 in ID x 2.000 in OD	91117A219seller's no. 91117A219McMaster-Carr	qty2buy 1 pack of 5	each$2.72$13.59 per 5	line$13.59used $5.44	1 alternate · 1 drop-inDROP-IN Fastenal 11103742 · $33.00 / 20 · Local DC Inventory; Direct Ship: Ships TOMORROWchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/washer_3_8	Oversized washer for 3/8 in, 1.25 in OD	91090A121seller's no. 91090A121McMaster-Carr	qty2buy 1 pack of 50	each$0.1424$7.12 per 50	line$7.12used $0.28	1 alternate · 1 drop-inDROP-IN Bolt Depot 2970 · $0.19 · no count shown; orderable (Add button, not marked 'Temporarily unavailable')checked 2026-10-27	☐	☐
☐	buy/washer_m12_thin	18-8 Stainless Steel Washer, Undersized, for M12 Screw Size, 13.0 mm ID, 20.0 mm OD	98689A118McMaster-CarrDelivers Wednesday 7-9 am (2026-10-27)	qty4buy 1 pack of 100	each$0.1371$13.71 per 100	line$13.71used $0.55	2 alternates · 2 drop-inDROP-IN Belmetric WF12X20SS · $0.20 · In stockchecked 2026-10-27DROP-IN Aspen Fasteners ME351-12 · Now: $179.53 · Contact Support for Availability Infochecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/washer_m3_large	18-8 Stainless Steel Oversized Washer, for M3 Screw, 3.2 mm ID, 9.0 mm OD, 0.7 mm to 0.9 mm Thick	91116A120McMaster-Carr	qty11buy 1 pack of 100	each$0.0481$4.81 per 100	line$4.81used $0.53	4 alternates · 4 drop-inDROP-IN Belmetric WFE3X9SS · $0.08 · in stockchecked 2026-10-27DROP-IN Aspen Fasteners ME380-3 · Now: $180.21 · Contact Support for Availability Infochecked 2026-10-27DROP-IN Belmetric WFE3X9SS · $0.08 · in stockchecked 2026-10-27DROP-IN Aspen Fasteners ME380-3 · Contact Support for Availability Infochecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/washer_m5	18-8 Stainless Steel General Purpose Washer, for M5 Screw Size, 5.3 mm ID, 10.0 mm OD	93475A240seller's no. 93475A240McMaster-Carr	qty6buy 1 pack of 100	each$0.0424$4.24 per 100	line$4.24used $0.25	2 alternates · 2 drop-inDROP-IN Belmetric WF5SS · $0.07 · in stockchecked 2026-10-27DROP-IN Bolt Depot 4515 · $0.05 · no count shown; orderable (Add button, not marked 'Temporarily unavailable')checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/washer_m6_din433	Washer, undersized (DIN 433), for M6 screw, 6.4 mm ID, 11.0 mm OD, 1.6 mm thick, 18-8 stainless steel	product page NO PART NUMBERMcMaster-Carr	qty4	eachNO PRICE	line—	NO ALTERNATE	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/washer_m6_large	18-8 Stainless Steel Oversized Washer, for M6 Screw, 6.4 mm ID, 18.0 mm OD, 1.4 mm to 1.8 mm Thick	91116A150seller's no. 91116A150McMaster-Carr	qty10buy 1 pack of 50	each$0.294$14.70 per 50	line$14.70used $2.94	2 alternates · 2 drop-inDROP-IN Bolt Depot 7315 · $0.12 · no count shown; orderable (Add button, not marked 'Temporarily unavailable')checked 2026-10-27DROP-IN Bolt Depot 7315 · $0.12 · no count shown; orderable (Add button, not marked 'Temporarily unavailable')checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/robstride-rs-04	Robstride RS-04	RS-04Seeed Studio ships from CN	qty2	each$255.00	line$510.00	8 alternates · 4 drop-inDROP-IN OpenELAB ROBSTRIDE04 · $297.99 (variant 'Motor (w/o Debug)'; Debug Board $24.99 extra) · Orderable; 8-15 business days; ships from China; duties includedchecked 2026-10-27DROP-IN AIFIT Lab ROBSTRIDE 04 Motor - Only Motor · $255.00 (Only Motor); $273.00 with USB-to-CAN module · Backorderedchecked 2026-10-27DROP-IN Robotics Center Robstride RS-04 (no SKU shown) · $295.00 · Backordered; 4-6 weekschecked 2026-10-27DROP-IN	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							rcdrone.top RobStride-04-1 · $319.00 (120nm Motor); $329.00 with CAN2USB debugger · Orderablechecked 2026-10-27CHANGES OpenELAB DM-J10010L-2EC · $321.99 · Orderable; ships from ChinaClosest substitute (Damiao DM-J10010L-2EC): same 40 N·m rated / 120 N·m peak and 200 rpm no-load at 48 V, but rated output speed is 100 rpm at 48 V vs the RS-04's 167 rpm. Round body ~120 mm dia x 53 mm (RS-04 is 120 x 120 x 56 mm), 1.37 kg vs 1.42 kg, 10:1 vs 9:1. The output flange and mounting-hole patterns differ (page		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							gives no hole pattern, so take it from Damiao's drawing), so the printed hip frame and the leg-link flange must be redrawn. Protocol changes from RobStride's CAN protocol to Damiao's CAN protocol (MIT, speed and position modes at 1 Mbps; UART at 921600 bps for setup), so the joint driver code must be rewritten. Supply is listed as 24-48 V (the RS-04 is 15-60 V), so check the pack's full-charge voltage. Connectors are the same types: XT30 power, GH1.25 CAN.checked 2026-10-27CHANGES AIFIT Lab DM-		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							J10010L-2EC (Include Driver / NOT Include) · $265.00 (motor with integrated driver); $280.00 with USB-to-CAN · BackorderedSame as the openelab DM-J10010L-2EC entry: new hip frame and flange, Damiao CAN protocol in place of RobStride's, 24-48 V supply, rated speed 100 rpm vs 167 rpm at 48 V.checked 2026-10-27CHANGES CubeMars AKE90-8 KV35 · $483.90 (actuator only; driver board not included)A stronger QDD (48 V, 55 N·m rated / 170 N·m peak, 120 rpm rated, 210 rpm no-load, 8:1), 1400 g, round 107.5 mm dia x		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							43.5 mm. CubeMars' comparison table lists 'Driver Board: No', so a separate driver (page downloads list an AK80-4830-1C-A4 driver; its price was not seen) must be bought, mounted and wired. That means a new hip frame and flange plus room for the driver, and CubeMars' CAN protocol in place of RobStride's in firmware. It costs about 1.9x the RS-04 before the driver.checked 2026-10-27CHANGES rcdrone.top RobStride-03-1 (RobStride 03) · $224.00 (Motor); $239.00 with USB-to-CAN ·		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							OrderableSame maker and same RobStride CAN protocol, so firmware changes the least (motor type and limits only). But it has half the torque: 20 N·m rated / 60 N·m peak, 195 rpm no-load. It is 106 x 106 x 56 mm and 880 g, with flange holes listed as '30.35 x 9 x 6 mm (M4)' and installation holes '98 x 8 x 8 mm (M4)' (the RS-04 lists M5 '35.86 x 9 x 9' and M4 '106 x 10 x 6'), so the hip frame and flange must be redrawn. Only usable if the leg torque budget allows it or another reduction stage is		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							added.checked 2026-10-27		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/odrive-botwheel	ODrive Robotics BotWheel (BW0124)	BotWheel (BW0124)ODrive Robotics shop	qty1buy 1 × 2-pack	each$134.50$269.00 per 2-pack	line$269.00used $134.50	3 alternates · 2 drop-inDROP-IN ODrive Robotics worldwide storefront BW0124_x2 · $269.00 per 2-pack (Single BW0124 $139.00 listed but unavailable) · Orderable; 2-packs onlychecked 2026-10-27DROP-IN ODrive Europe BW0124_x2 · EUR 269.00 per pair · In stock (40)checked 2026-10-27CHANGES AIFIT Lab ZLLG65ASM250-4096 V2.0 (Single Shaft / Not Include Driver) · $120.00 (motor only); $260.00 with ZLAC8015 driver; $390.00 two motors with ZLAC8015D · BackorderedTh	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							e mount changes completely. It has a single-sided cantilever axle with an M16x1.5 thread and 12 mm flats (43 mm long), held by a nut through a slotted plate, instead of the BotWheel flange taking 4 x M6 on a 36 mm circle, so the wheel mount at the foot of the leg must be redrawn. Tyre is 173 ±2 mm dia x 44 mm wide (BotWheel 171 mm), overall width 118.2 mm. Mass is 3.75 kg vs 2.2 kg, i.e. +1.55 kg at each leg tip, which eats into the leg torque budget and needs the balance		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							controller retuned. It has a 4096-line encoder and 15 pole pairs (BotWheel: hall + 3200 CPR), and ZLTECH power and signal connectors instead of JST PUD, so the ODrive harness must be re-terminated and its encoder and pole-pair settings changed; the alternative is ZLTECH's ZLAC8015D driver, which needs different firmware. It is 24 V, 6 N·m rated / 18 N·m peak, 160 rpm rated, IP65.checked 2026-10-27		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/lumenier-razor-pro-f4-45a-am32	Lumenier Razor Pro F4 45A 2-6S AM32 single ESC (the flywheel ESC's backup)	Razor Pro F4 45A 2-6S AM32 ESC (single)seller's no. 22300 (UPC 724376363650)RaceDayQuadsavailable: true (Shopify; no count exposed) (2026-10-27)	qty2	each$31.49	line$62.98	1 alternate · 1 drop-inDROP-IN GetFPV not read · $31.49 · unknown (page returns 403 to automated reads)checked 2026-10-27	☐	☐
☐	buy/heatshrink_kit	Heat-shrink assortment	Heat Shrink Tube Set (530PC)seller's no. Heat Shrink Tube Set (530PC)RaceDayQuads	qty1	each$9.99	line$9.99	1 alternate · 1 drop-inDROP-IN SparkFun PRT-09353 · $10.95checked 2026-10-27	☐	☐
☐	buy/xt60_pair	XT60H connector	Amass XT60H (male + female)seller's no. Amass XT60H (male + female)RaceDayQuads	qty4	each$0.990	line$3.96	1 alternate · 1 drop-inDROP-IN Amazon ASIN B0863HJZHB (XT60H pairs) · $1.80checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/xt30_male	XT30 connector, male	Amass XT30 (male)seller's no. Amass XT30 (male)RaceDayQuads	qty3	each$0.990	line$2.97	1 alternate · 1 drop-inDROP-IN GetFPV 5071 (XT30 Power Connectors, 5 Pair) · $3.00 / 5 · In Stockchecked 2026-10-27	☐	☐
☐	buy/tool-am32-usb-linker	USB linker / ESC programming card for AM32 and BLHeli_32 ESCs (tool)	VXB0FNQW93SR (Shopify variant 48952943968491)seller's no. VXB0FNQW93SR (Shopify variant 48952943968491)VXBavailable: true (2026-10-27)	qty1	each$49.99	line$49.99	1 alternate · 1 drop-inDROP-IN Amazon ASIN B0FNQW93SR · not shown to this readerchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/vxb-6810-2rs	6810-2RS Bearing 50x65x7mm - Sealed Ball Bearing - Thin Section (chrome steel)	Kit7458 (6810-2RS)VXB Bearings	qty1	each$24.99	line$24.99	2 alternates · 2 drop-inDROP-IN McMaster-Carr 6656K229 · $33.29 · Delivers tomorrow 7-9 amchecked 2026-10-27DROP-IN Bearings Direct 6810-2RS · $29.19 · In stock - 1 available; usually ships in 24 hourschecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/feetech-sts3215-c018-12v	Feetech (Waveshare ST3215) STS3215 / ST3215 12 V 30 kg-cm serial bus servo	STS3215 / ST3215 12 V 30 kg-cm serial bus servoseller's no. 22414Waveshareschema.org availability InStock; SKU 22414 'unsaleable': false (no count shown) (2026-10-27)	qty3	each$21.99	line$65.97	13 alternates · 11 drop-inDROP-IN OpenELAB Feetech-STS3215-C018 · $27.99 · Orderable; ships from Chinachecked 2026-10-27DROP-IN AIFIT Lab ST-3215-C018 (12V30KG) · $26.00 · Backorderedchecked 2026-10-27CHANGES OpenELAB Feetech-STS3250-C002 · $73.99 · OrderableSame Feetech STS serial-bus protocol and 3-pin TTL connector, so firmware stays the same apart from limits. It is stronger (50 kg·cm stall at 12 V, 345:1, 6-12 V) and costs about 2.6x. The page gives no case size or	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							spline, and it says 'a matching TTL interface alone does not establish mechanical compatibility', so check the case and horn against the 45.2 x 24.7 x 35 mm STS3215 before reusing the turret brackets.checked 2026-10-27CHANGES ROBOTIS America 902-0135-000 (DYNAMIXEL XL430-W250-T) · $27.50 · OrderableThe case is 28.5 x 46.5 x 34 mm with a different horn and mounting holes, so the pan/tilt brackets must be redrawn. It speaks DYNAMIXEL protocol over TTL instead of Feetech STS, so		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							the servo driver/SDK and ID setup change. It is weaker: the page gives an estimated rated torque of 0.28 N·m, against about 0.98 N·m (10 kg·cm) rated for the STS3215-C018. Only worth it if the turret load is light.checked 2026-10-27DROP-IN Amazon ASIN B0CFY52HVV · $30.99 · In Stock (quantity selector to 20+)checked 2026-10-27DROP-IN Amazon ASIN B0B6P9271Q · $30.90 · Only 1 left in stock - order soonchecked 2026-10-27DROP-IN Amazon ASIN B0FMXVR77Y · $48.99 / 2 · orderable: Add		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							to Cart with a quantity selector (1-8+); the availability line is blankchecked 2026-10-27DROP-IN Amazon ASIN B0CFY52HVV · $30.99 · In Stock (quantity selector to 20+)checked 2026-10-27DROP-IN Amazon ASIN B0B6P9271Q · $30.90 · Only 1 left in stock - order soonchecked 2026-10-27DROP-IN Amazon ASIN B0FMXVR77Y · $48.99 / 2 · orderable: Add to Cart with a quantity selector (1-8+); the availability line is blankchecked 2026-10-27DROP-IN Amazon ASIN B0CFY52HVV · $30.99 · In		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							Stock (quantity selector to 20+)checked 2026-10-27DROP-IN Amazon ASIN B0B6P9271Q · $30.90 · Only 1 left in stock - order soonchecked 2026-10-27DROP-IN Amazon ASIN B0FMXVR77Y · $48.99 / 2 · orderable: Add to Cart with a quantity selector (1-8+); the availability line is blankchecked 2026-10-27		
☐	buy/waveshare-bus-servo-adapter-a	USB / UART adapter for Feetech serial bus servos	Bus Servo Adapter (A), SKU 25514seller's no. 25514 (MPN 'Bus Servo Adapter (A)')Waveshareschema.org availability InStock (2026-10-27)	qty1	each$4.99	line$4.99	1 alternate · 1 drop-inDROP-IN Amazon ASIN B0CJ6TP3TP · $12.99 · In Stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/repeat-robotics-am32-35a-esc	Repeat Robotics (sold by Palm Beach Bots) AM32 Brushless Drive ESC 35 A (bi-directional)	AM32 Brushless Drive ESC 35 A (bi-directional)Palm Beach Bots	qty2	each$24.99	line$49.98	2 alternates · 1 drop-inDROP-IN Repeat Robotics RR-03-3340 (AM32 35a Single Drive ESC) · $25.00 · Orderablechecked 2026-10-27CHANGES RaceDayQuads 22300 (Lumenier Razor Pro F4 45A 2-6S AM32 ESC) · $31.49 · OrderableAt 12 x 28 x 5 mm (3 g without motor wires) it is 2-3 mm longer than the Repeat board's 25-26 x 13 mm, so check the ESC pocket. It is rated 45 A continuous / 60 A burst, 2-6S, no BEC. The page lists DShot1200, Proshot, Oneshot125, Oneshot42 and Multishot input with auto-detect but does	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							not list standard servo PWM, so confirm the controller's output protocol. It is not pre-tuned: set direction, bidirectional mode and throttle limits in the AM32 configurator. It ships with a 220 µF 35 V capacitor that needs a place in the wiring.checked 2026-10-27		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/xt60e_f_panel	XT60 female panel socket (the motors' outlet)	Amass XT60E-Fseller's no. Amass XT60E-FJameco Electronics	qty1	each$0.950	line$0.95	3 alternates · 3 drop-inDROP-IN Maplin XT60E-F (Maplin SKU 486711) · Out of stockchecked 2026-10-27DROP-IN Electrokit XT60E-F (Electrokit art. 4103 7928) · Out of stockchecked 2026-10-27DROP-IN LCSC XT60E-M.G.Y (Amass male panel-mount version; LCSC C3040679) · $0.61 · In stock (3,091)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/bearing_61812	Steel Ball Bearing, Steel, Shielded, Trade Number 6812-2Z, ABEC Rating 1	6812-2RSBearings DirectIn stock (5) (2026-10-27)	qty4	each$38.58	line$154.32	3 alternates · 3 drop-inDROP-IN McMaster-Carr (the catalogue's) 6656K236 · $69.11 eachchecked 2026-10-26DROP-IN VXB Bearings Kit7454 · $49.99 · In stockchecked 2026-10-27DROP-IN McMaster-Carr 6656K148 · $64.92 each · Delivers tomorrow 7-9 amOpen (no shields): the same 60 x 78 x 10 seat, but nothing keeps the grease in or grit out. Fine in the hip frame; at the apex, next to the wheel, prefer the shielded one.checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/bearing_mr126	Corrosion-Resistant 440C Stainless Steel Ball Bearing, Shielded, Trade No. 126-2Z, for 6 mm Shaft Diameter	MR126-2RSBearings DirectIn stock - 646 available (2026-10-27)	qty1	each$4.56	line$4.56	3 alternates · 2 drop-inDROP-IN McMaster-Carr (the catalogue's) 7804K112 · $8.89 eachchecked 2026-10-27DROP-IN VXB Bearings MR126-2RS · $7.77 · In stockchecked 2026-10-27CHANGES McMaster-Carr 7804K142 · $9.08 each · Delivers tomorrow 7-9 amThe flanged 126-2Z (flange Ø13.6 x 0.8): the yoke's Ø12 x 4 pocket needs a Ø14 x 0.8 counterbore, or the flange stands 0.8 mm proud of the arm's inner face.checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/bearing_6301	Deep groove ball bearing 6301-2RSH (12 x 37 x 12, sealed both sides); McMaster-Carr 5972K86 is the same trade number (6301-2RS)	6301-2RSHSKF distributor	qty4	each$12.53	line$50.12	NO ALTERNATE	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/dewalt-dcb609	DeWalt DCB609 FLEXVOLT 20V/60V MAX 9.0Ah	DCB609 FLEXVOLT 20V/60V MAX 9.0AhThe Power Tool Store	qty3	each$259.00	line$777.00	1 alternate · 1 drop-inDROP-IN Ace Tool product page · $259.00 · listedchecked 2026-10-27	☐	☐
☐	buy/bracket_corner_8	Corner bracket, 8 series, with 2 x M8 screws + 2 drop-in spring nuts	HBLTS8-C-SEPseller's no. HBLTS8-C-SEPMISUMIShip Date Same Day (2026-10-27)	qty24	each$4.42	line$106.08	2 alternates · 2 drop-inDROP-IN MISUMI HBLTS8-C-SEC · $5.24checked 2026-10-27DROP-IN 80/20 Inc. 40-4376 + 3 x 75-3422 · $10.90checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/extrusion_8080_1200	Aluminium extrusion 8080, 8 series (10 mm slot), cut to 1200 mm	HFS8-8080-1200seller's no. HFS8-8080-1200MISUMIShip Date Wed. Oct 28, 2026 (2026-10-27)	qty1	each$97.44	line$97.44	1 alternate · 1 drop-inDROP-IN 80/20 Inc. 40-8080 cut to 1200 mm · $151.73checked 2026-10-27	☐	☐
☐	buy/extrusion_4040_620	Aluminium extrusion 4040, 8 series (10 mm slot), cut to 620 mm	HFS8-4040-620seller's no. HFS8-4040-620MISUMIShip Date Wed. Oct 28, 2026 (2026-10-27)	qty4	each$19.09	line$76.36	1 alternate · 1 drop-inDROP-IN 80/20 Inc. 40-4040 cut to 620 mm · $30.34checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/extrusion_4040_900	Aluminium extrusion 4040, 8 series (10 mm slot), cut to 900 mm	HFS8-4040-900seller's no. HFS8-4040-900MISUMIShip Date Wed. Oct 28, 2026 (2026-10-27)	qty2	each$27.72	line$55.44	1 alternate · 1 drop-inDROP-IN 80/20 Inc. 40-4040 cut to 900 mm · $42.69checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/bracket_free_angle_8	Free-angle bracket for 40 mm 8-series profile, with its mounting screws and nuts	HBLTBA40-SET1seller's no. HBLTBA40-SET1MISUMIShip Date Same Day (2026-10-27)	qty4	each$8.78	line$35.12	2 alternates · 2 drop-inDROP-IN MISUMI HBLTBBN40-SET · $10.15checked 2026-10-27DROP-IN 80/20 Inc. 40-2085 hinge · $14.34checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/extrusion_4080_620	Aluminium extrusion 4080, 8 series (10 mm slot), cut to 620 mm	HFS8-4080-620seller's no. HFS8-4080-620MISUMIShip Date Wed. Oct 28, 2026 (2026-10-27)	qty1	each$34.65	line$34.65	1 alternate · 1 drop-inDROP-IN 80/20 Inc. 40-4080 cut to 620 mm · $54.69checked 2026-10-27	☐	☐
☐	buy/nut_m4_dropin	Drop-in nut, 8 series, M4	HNTA8-4seller's no. HNTA8-4MISUMIShip Date Same Day (2026-10-27)	qty36	each$0.900	line$32.40	1 alternate · 1 drop-inDROP-IN MISUMI PACK-HNTA8-4 (100) · $62.00 / 100checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/extrusion_4040_720	Aluminium extrusion 4040, 8 series (10 mm slot), cut to 720 mm	HFS8-4040-720seller's no. HFS8-4040-720MISUMIShip Date Wed. Oct 28, 2026 (2026-10-27)	qty1	each$22.17	line$22.17	1 alternate · 1 drop-inDROP-IN 80/20 Inc. 40-4040 cut to 720 mm · $34.75checked 2026-10-27	☐	☐
☐	buy/nut_m8_dropin	Drop-in nut, 8 series, M8	HNTA8-8seller's no. HNTA8-8MISUMIShip Date Same Day (2026-10-27)	qty16	each$0.900	line$14.40	1 alternate · 1 drop-inDROP-IN 80/20 Inc. 75-3422 (M8 x 16 BHSCS + slide-in nut) · $1.14checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/nut_m6_dropin	Drop-in nut, 8 series, M6	HNTA8-6seller's no. HNTA8-6MISUMIShip Date Same Day (2026-10-27)	qty12	each$0.900	line$10.80	1 alternate · 1 drop-inDROP-IN Fastenal 0961750 (80/20 Inc 13129) · $2.83 · Available Inventory; ships from the manufacturer within 9 dayschecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/rail_hgr15_kit	HGR15 rail 1000 mm x 2 + HGH15CA block x 4 (one pack)	ASIN B0GJZCLXWJ (1000mm / 2)seller's no. ASIN B0GJZCLXWJ (1000mm / 2)AmazonOnly 12 left in stock (2026-10-27)	qty1	each$69.99	line$69.99	3 alternates · 3 drop-inDROP-IN Amazon (Coavoo Transmission) ASIN B0BMWQ5J7T · $72.99 · Only 3 left in stockchecked 2026-10-27DROP-IN VXB Bearings VXB0BMWQ5J7T · $159.95checked 2026-10-27DROP-IN Amazon (Mycncshop) ASIN B087ZCH3BW (FBT HGR15CA-L1000mm) · $151.00checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/terrafirma-dewalt-20v-adapter-10awg	Terrafirma Technology DeWalt 20 V battery adapter, 10 AWG leads (1 ft), 30 A stated, no fuse or switch, 4-pack	4pack 10AWG Adapter for DeWALT 20V (Amazon B07HYJ628S)Amazonlisted (2026-10-27)	qty1buy 1 pack of 4	each$16.00$64.00 per 4	line$64.00used $16.00	1 alternate · 1 drop-inDROP-IN Amazon (Terrafirma, 12 AWG single) product page · $19.00 · listed; 12 AWG, M4/#6 mounting holes, current not statedchecked 2026-10-27	☐	☐
☐	buy/rubber_mat	Rubber mat 3 x 4 ft x 5/16 in, 2-pack	ASIN B0F8RCXB7Dseller's no. ASIN B0F8RCXB7DAmazon	qty1	each$52.00	line$52.00	1 alternate · 1 drop-inDROP-IN Amazon (uxcell) ASIN B0DP9TXB59 (M8 levelling feet, 8) · $34.00 / 8checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/sling_1in	Lift-All Type 5 endless polyester sling 1 in x 5 ft (2400 lb vertical)	ASIN B0026GIUMGseller's no. ASIN B0026GIUMGAmazon	qty2	each$17.08	line$34.16	1 alternate · 1 drop-inDROP-IN Amazon (Kivomot) ASIN B0FRRVHXTR (2-pack, eye-eye 6.5 ft) · $12.58 / 2checked 2026-10-27	☐	☐
☐	buy/carabiner	Screwgate carabiner, 25 kN, 2-pack	ASIN B07D73RRH5 (2-pack)seller's no. ASIN B07D73RRH5 (2-pack)Amazon	qty4buy 2 packs of 2	each$8.50$16.99 per 2	line$33.98	1 alternate · 1 drop-inDROP-IN Amazon (STURME) ASIN B085HLB374 (2-pack) · $16.99 / 2checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/dock_adapter_20v	DeWalt 20 V MAX dock adapter, 12 AWG leads, switch and 30 A blade fuse	ASIN B0B9NPZM3Mseller's no. ASIN B0B9NPZM3MAmazon In Stock (2026-10-27)	qty2	each$9.99	line$19.98	1 alternate · 1 drop-inDROP-IN Amazon (Ensayeer, sold by Unsaidtech) ASIN B09J4R4QZ6 (2-pack) · $22.00 / 2 · In Stockchecked 2026-10-27	☐	☐
☐	buy/xt90s_pair	Amass XT90-S anti-spark connector, male + female, 5 pairs	XT90-SSky RC Hobby on AmazonIn Stock (ships from Amazon) (2026-10-27)	qty2buy 1 pack of 5	each$2.60$12.98 per 5	line$12.98used $5.19	1 alternate · 1 drop-inDROP-IN FLYRC on Amazon product page · $14.99 / 3 · In Stock; 3 pairs with 150 mm 10 AWG pigtailschecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/load_cells_4x50	4 x 50 kg half-bridge load cells + HX711	ASIN B079FTXR7Yseller's no. ASIN B079FTXR7YAmazonOnly 10 left (2026-10-27)	qty1	each$7.90	line$7.90	1 alternate · 1 drop-inDROP-IN VXB VXB07X87GYLQ · $39.99checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/traco-tep-150-4812wi	TRACO TEP 150-4812WI DC-DC converter, 18-75 V in (4:1), 12 V 12.5 A (150 W), isolated 2,250 V DC, chassis mount, screw terminals	TEP 150-4812WINewark4 (as an aggregator, electronicsdatasheets.com, lists it: confirm on the order) (2026-10-27)	qty1	each$275.99	line$275.99	1 alternate · 1 drop-inDROP-IN Farnell product page · $222.65 · listed (via the same aggregator)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/tvs-p6ke51a	TVS diode 600 W unidirectional, 43.6 V standoff, 70.1 V clamp, DO-15 (P6KE51A)	P6KE51Aseller's no. 99AC4212Newark1476 (2026-10-27)	qty3	each$1.28	line$3.84	1 alternate · 1 drop-inDROP-IN TME P6KE51A-DIO · $0.47 · 4628checked 2026-10-27	☐	☐
☐	buy/yageo-mfr-25fbf52-10r	Resistor 10.0 R 1 % 1/4 W metal film (Yageo MFR-25FBF52)	MFR-25FBF52-10Rseller's no. 49AK5972Newark9608 (2026-10-27)	qty2	each$0.100	line$0.20	1 alternate · 1 drop-inDROP-IN Quest Components product page · $0.07 · 88checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/remington-18ptfestr-25ft	Hook-up wire 18 AWG stranded PTFE, 600 V, 25 ft	18PTFESTRBLA25 (black) / 18PTFESTRRED25 (red)Digi-Key	qty1buy 1 pack of 25	each$2.58$64.56 per 25	line$64.56used $2.58	1 alternate · 1 drop-inDROP-IN Mouser 602-6825-100-02 · $120.87 / 100 · 2 (Mouser, via oemstrade feed today)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/tdk-lambda-i6a4w020a033v-001-r	i6A4W 250 W non-isolated buck, 9-53 V in, 3.3-15 V out by one trim resistor, 20 A (1/16 brick, horizontal through-hole)	i6A4W020A033V-001-Rseller's no. I6A4W020A033V-001-RDigi-Key1873 (2026-10-27)	qty1	each$47.46	line$47.46	1 alternate · 1 drop-inDROP-IN Newark 81Y1493 · $40.32 · 36 in one snapshot, back-order in anotherchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/quarton-vlm-650-03-lpt	Red dot laser module 650 nm, < 1 mW (Class 2), 2.6-6 V, Ø7 x 22.5 mm	VLM-650-03 LPTseller's no. VLM-650-03-LPT-NDDigi-Key966 (authorized; Octopart data updated 2026-10-26 12:48 UTC) (2026-10-27)	qty1	each$25.03	line$25.03	3 alternates · 1 drop-inCHANGES Digi-Key Quarton VLM-650-28 LPT · 63 (snippet)check its diameter and length against the Ø7 bore before orderingchecked 2026-10-27CHANGES Digi-Key Quarton VLM-520-73 LPT (green, Class 2) · $43.49 · 189 (Octopart 2026-10-26)Ø7 x 26.1: 5 mm longer than the red one; the mount holds it as drawn for v1checked 2026-10-26DROP-IN Digi-Key VLM-635-03LPT-ND · $28.91 · 36 (authorized; Octopart data updated 2026-10-26 12:48 UTC)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/infineon-irlz44npbf	Logic-level N-MOSFET 55 V 47 A, TO-220	IRLZ44NPBFseller's no. IRLZ44NPBF-NDDigi-Key	qty6	each$1.80	line$10.80	1 alternate · 1 drop-inDROP-IN Mouser 942-IRLZ44NPBF · $1.80 · 77849checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/te-dt06-12sa-kit	CONN PLUG HSG 12POS - DT 12-way plug (socket side), key A	DT06-12SA (+ W12S wedge + 12 x 0462-201-16141 sockets)Digi-Key	qty1	each$8.50	line$8.50	1 alternate · 1 drop-inDROP-IN Chief Enterprises AT06-12SA plug + AW12S wedge + 12 x AT62-201-16141 sockets (Amphenol Sine Systems) · AT06-12SA $3.22 (1-9); AW12S $0.31 (1-9); AT62-201-16141 $0.58 (1-9). Kit = $3.22 + $0.31 + 12 x $0.58 = $10.49 · OrderableNone if paired with the Amphenol AT04-12PA-L012 receptacle (Amphenol names AT04-12PA as the mate). Mating with the TE DT04-12PA-L012 relies on AT/DT intermating, which no page I read states.checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/stop_relay_dpdt	Omron G2RL-2 DC12 power relay, DPDT, 12 V coil, 8 A at 30 V DC (the stop relay: the turret's safety loop and its 24 V branch)	G2RL-2-DC12DigiKey1,078 immediate (search result, 2026-10-27; the live page to confirm) (2026-10-27)	qty2	each$4.14	line$8.28	1 alternate · 1 drop-inDROP-IN LCSC product page · listed as C61458 (price shown on the live page)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/fuse_holder_5x20	5 x 20 mm panel fuse holder	Littelfuse 03455LS1Hseller's no. Littelfuse 03455LS1HDigi-Key	qty1	each$7.16	line$7.16	2 alternates · 2 drop-inDROP-IN LCSC 03455LS1HX020 (Littelfuse; LCSC C206937) · $1.40 · Out of stockchecked 2026-10-27DROP-IN Schurter 0031.1756 (FEU = body 0031.1673 + fingergrip cap 0031.1741, 5x20 mm, solder, 90 deg)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/fuse-holder-5x20-panel	FUSE HLDR CART 250V 10A PNL MNT-345 series shock-safe panel holder, 5 x 20 mm, low-profile body, screwdriver-slot knob, 3/16 in right-	03455LS1HDigi-Key	qty1	each$7.16	line$7.16	2 alternates · none drop-inCHANGES LCSC 03455LS1HX020 (Littelfuse; LCSC C206937) · $1.3968 (1+), $1.1671 (10+), $1.0407 (30+), $0.8973 (100+), $0.8341 (500+), $0.8048 (1000+) · Out of stockSame 345-series holder as the primary (the page does not explain the X020 suffix), so it needs the same 12.70-12.83 mm hole with a flat, not a 12.6 mm round hole.checked 2026-10-27CHANGES Schurter 0031.1756 (FEU = body 0031.1673 + fingergrip cap 0031.1741, 5x20 mm, solder, 90 deg)M12.7 x	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
		angle dual-purpose solder/QC terminals					1.5 thread (nut SW14). The datasheet's mounting-hole figures are 12.7 (+0.1/0) and 13 (-0.3) mm round, with flat options 12.2 (-0.13) and 11.7 (-0.13) mm (read as text; check the drawing). The 12.6 mm hole is too small; open it to about 12.8-13.0 mm. Panel thickness max 7 mm, so the 4 mm wall is fine. Caps are ordered separately unless the complete code is used.checked 2026-10-27		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/te-0460-202-16141	DT size 16 pin contact, nickel, 14-16 AWG	0460-202-16141seller's no. 1734-1328-NDDigi-Key	qty12	each$0.450	line$5.40	1 alternate · 1 drop-inDROP-IN Mouser 571-0460-202-16141 · $0.45 · 267161checked 2026-10-27	☐	☐
☐	buy/littelfuse-0477005-mxp	5 x 20 mm time-lag fuse, 5 A, 500 V AC / 400 V DC (477 series)	Littelfuse 0477005.MXPseller's no. Littelfuse 0477005.MXPDigi-Key	qty4	each$1.00	line$4.00	2 alternates · 2 drop-inDROP-IN Mouser 576-0477005.MXP (Littelfuse 0477005.MXP) · $2.18 · 8590 in stock (Mouser's figure as listed on findchips.com today)checked 2026-10-27DROP-IN Digi-Key product pagechecked ?	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/molex-43031-0001	Micro-Fit 3.0 male crimp terminal, 20-24 AWG, tin	43031-0001seller's no. WM1841CT-NDDigi-Key	qty40	each$0.085	line$3.40	1 alternate · 1 drop-inDROP-IN Mouser 538-43031-0001-CT · $8.50 / 100 · 5000checked 2026-10-27	☐	☐
☐	buy/panasonic-eeu-fr1j471b	Aluminium electrolytic, 470 uF, 63 V, low ESR (FR), 12.5 x 25 mm radial	EEU-FR1J471Bseller's no. P15300CT-NDDigi-Key~20,107 (findchips row, undated) (2026-10-27)	qty2	each$1.54	line$3.08	1 alternate · 1 drop-inDROP-IN Newark 62W8085 · $2.13 · 10970checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/allegro-a1120lua-t	Unipolar hall-effect switch, 3-24 V, SIP-3 (UA)	A1120LUA-Tseller's no. 620-1330-NDDigi-Key22190 (2026-10-27)	qty1	each$1.79	line$1.79	1 alternate · 1 drop-inDROP-IN Mouser 250-A1120LUA-T · $1.79 · 2262checked 2026-10-27	☐	☐
☐	buy/molex-43020-1200	Micro-Fit 3.0 plug housing, dual row, 12 circuits	43020-1200seller's no. WM1782-NDDigi-Key	qty2	each$0.820	line$1.64	1 alternate · 1 drop-inDROP-IN Mouser 538-43020-1200 · $0.82 · 13814checked 2026-10-27	☐	☐
☐	buy/molex-43025-1200	Micro-Fit 3.0 receptacle housing, dual row, 12 circuits	43025-1200seller's no. WM1788-NDDigi-Key	qty2	each$0.810	line$1.62	1 alternate · 1 drop-inDROP-IN Mouser 538-43025-1200 · $0.81 · 75541checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/vishay-4n35	Optocoupler, phototransistor output, DIP-6	4N35seller's no. 4N35-NDDigi-Key	qty1	each$1.58	line$1.58	1 alternate · 1 drop-inDROP-IN Mouser 859-4N35 · $0.64 · 9390checked 2026-10-27	☐	☐
☐	buy/bourns-mf-r300	Resettable PTC fuse, 3 A hold, 30 V	MF-R300seller's no. MF-R300-NDDigi-Key	qty3	each$0.500	line$1.50	1 alternate · 1 drop-inDROP-IN Mouser 652-MFR300 · $0.50 · 5353checked 2026-10-27	☐	☐
☐	buy/vishay-teft4300	Silicon NPN phototransistor, T-1 (3 mm)	TEFT4300seller's no. 751-1041-NDDigi-Key	qty2	each$0.700	line$1.40	1 alternate · 1 drop-inDROP-IN Mouser 782-TEFT4300 · $0.70 · 2446checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/molex-43030-0038	Micro-Fit 3.0 female crimp terminal, 18 AWG, tin, 8.5 A	43030-0038seller's no. WM13070CT-NDDigi-Key	qty8	each$0.160	line$1.28	1 alternate · 1 drop-inDROP-IN Mouser 538-43030-0038-LP · $0.16 · 79499checked 2026-10-27	☐	☐
☐	buy/molex-43031-0021	Micro-Fit 3.0 male crimp terminal, 18 AWG, tin, 8.5 A	43031-0021seller's no. WM22521CT-NDDigi-Key	qty8	each$0.160	line$1.28	1 alternate · 1 drop-inDROP-IN Mouser 538-43031-0021-LP · $0.16 · 9188checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/onsemi-2n7000	N-channel MOSFET 60 V 200 mA, TO-92	2N7000seller's no. 2N7000FS-NDDigi-Key	qty1	each$1.23	line$1.23	1 alternate · 1 drop-inDROP-IN Mouser 512-2N7000 · $1.23 · 95079checked 2026-10-27	☐	☐
☐	buy/ti-sn74hc74n	Dual D flip-flop, DIP-14	SN74HC74Nseller's no. 296-1602-5-NDDigi-Key9538 (2026-10-27)	qty1	each$1.13	line$1.13	1 alternate · 1 drop-inDROP-IN Mouser 595-SN74HC74N · $1.11 · 10808checked 2026-10-27	☐	☐
☐	buy/molex-43030-0001	Micro-Fit 3.0 female crimp terminal, 20-24 AWG, tin	43030-0001seller's no. WM1837CT-NDDigi-Key	qty14	each$0.075	line$1.05	1 alternate · 1 drop-inDROP-IN Mouser 538-43030-0001-CT · $7.50 / 100 · 257100checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/molex-43020-1000	Micro-Fit 3.0 plug housing, dual row, 10 circuits	43020-1000seller's no. WM1781-NDDigi-Key	qty1	each$0.970	line$0.97	1 alternate · 1 drop-inDROP-IN Mouser 538-43020-1000 · $0.97 · 20449checked 2026-10-27	☐	☐
☐	buy/vishay-tsal4400	IR emitting diode 940 nm, T-1 (3 mm)	TSAL4400seller's no. 751-1201-NDDigi-Key	qty2	each$0.480	line$0.96	1 alternate · 1 drop-inDROP-IN Mouser 782-TSAL4400 · $0.48 · 18862checked 2026-10-27	☐	☐
☐	buy/ti-sn74ahct125n	Quad buffer, 3-state, TTL inputs, DIP-14	SN74AHCT125Nseller's no. 296-4655-5-NDDigi-Key1674 (2026-10-27)	qty1	each$0.950	line$0.95	1 alternate · 1 drop-inDROP-IN Mouser 595-SN74AHCT125N · $0.98 · 3136checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/te-w12p	DT wedgelock for the 12-way receptacle	W12Pseller's no. 1734-1034-NDDigi-Key	qty1	each$0.880	line$0.88	1 alternate · 1 drop-inDROP-IN Mouser 571-W12P-P004 · $0.88 · 2620checked 2026-10-27	☐	☐
☐	buy/ti-sn74hc08n	Quad 2-input AND, DIP-14	SN74HC08Nseller's no. 296-1570-5-NDDigi-Key22274 (2026-10-27)	qty1	each$0.790	line$0.79	1 alternate · 1 drop-inDROP-IN Mouser 595-SN74HC08N · $0.78 · 8674checked 2026-10-27	☐	☐
☐	buy/molex-43025-1000	Micro-Fit 3.0 receptacle housing, dual row, 10 circuits	43025-1000seller's no. WM1787-NDDigi-Key	qty1	each$0.750	line$0.75	1 alternate · 1 drop-inDROP-IN Mouser 538-43025-1000 · $0.75 · 131360checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/ti-ne555p	Precision timer, DIP-8	NE555Pseller's no. 296-NE555P-NDDigi-Key28579 (2026-10-27)	qty1	each$0.590	line$0.59	1 alternate · 1 drop-inDROP-IN Mouser 595-NE555P · $0.59 · 61909checked 2026-10-27	☐	☐
☐	buy/ti-lm393p	Dual comparator, DIP-8	LM393Pseller's no. 296-1398-5-NDDigi-Key8592 (2026-10-27)	qty1	each$0.530	line$0.53	1 alternate · 1 drop-inDROP-IN Mouser 595-LM393P · $0.53 · 23356checked 2026-10-27	☐	☐
☐	buy/molex-43025-0600	Micro-Fit 3.0 receptacle housing, dual row, 6 circuits	43025-0600seller's no. WM1785-NDDigi-Key	qty1	each$0.510	line$0.51	1 alternate · 1 drop-inDROP-IN Mouser 538-43025-0600 · $0.51 · 207720checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/bourns-mf-r250	PTC resettable fuse, 30 V, 2.5 A hold, 5.0 A trip, radial	MF-R250seller's no. MF-R250-NDDigi-Key	qty1	each$0.490	line$0.49	1 alternate · 1 drop-inDROP-IN Mouser 652-MFR250 · $0.49 · 8520checked 2026-10-27	☐	☐
☐	buy/molex-43020-0600	Micro-Fit 3.0 plug housing, dual row, 6 circuits	43020-0600seller's no. WM1779-NDDigi-Key	qty1	each$0.400	line$0.40	1 alternate · 1 drop-inDROP-IN Mouser 538-43020-0600 · $0.40 · 12657checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/caldwell-ballistic-precision-chronograph-bt	Ballistic Precision chronograph with Bluetooth (optical)	4001147seller's no. SKU 4001147CaldwellAvailability: In Stock (2026-10-27)	qty1	each$168.99	line$168.99	2 alternates · 2 drop-inDROP-IN Competition Electronics CEI-3820 · $139.95 · available: truechecked 2026-10-27DROP-IN Graf & Sons CE3820 (Shopify variant 46624058671235) · $136.99 · available: truechecked 2026-10-27	☐	☐
☐	buy/pololu-d36v50f12-4095	12V, 4.5A Step-Down Voltage Regulator D36V50F12	4095 (D36V50F12)Pololu	qty2	each$39.95	line$79.90	1 alternate · 1 drop-inDROP-IN Pololu 5577 · $47.95 · 220checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/buck_5v	Pololu 5 V 3.2 A step-down regulator D36V28F5 (from the 12 V rail)	D36V28F5 (#3782)PololuRationed (active and preferred) (2026-10-27)	qty1	each$23.95	line$23.95	1 alternate · 1 drop-inDROP-IN Pololu product page · $26.95 · Rationed; D24V22F5 5 V 2.5 A, 5.3-36 V inchecked 2026-10-27	☐	☐
☐	buy/pololu-d36v28f5-3782	5V, 3.2A Step-Down Voltage Regulator D36V28F5	3782 (D36V28F5)Pololu	qty1	each$23.95	line$23.95	1 alternate · none drop-inCHANGES Pololu 2858 · $26.95 · 141D24V22F5: 2.5 A against 3.2 A, input at most 36 V (fine on 24 V); a different footprint on the level-1 boardchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/radio_stop_relay	Pololu RC switch with relay (the radio stop: its dry contact is the STOP loop to the MCU; D37 calls this the simplest non-rated version)	2804Pololulisted (the body plan, read 2026-10-27) (2026-10-27)	qty1	each$19.95	line$19.95	1 alternate · 1 drop-inDROP-IN Pololu product page · the digital-output version (no relay): backup through a MOSFET on the MCU boardchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/pololu-ul1007-24awg-60ft	Stranded hook-up wire, 24 AWG UL1007, 60 ft spool	2630 (black), 2632 (red), 2634 (yellow), 2639 (white)seller's no. 2630PololuInStock (page data-available-stock=441); category page showed some colours as out of stock/backorder (2026-10-27)	qty4buy 1 pack of 60	each$0.1825$10.95 per 60	line$10.95used $0.73	1 alternate · 1 drop-inDROP-IN Digi-Key product page · $31.63 / 6checked 2026-10-27	☐	☐
☐	buy/pololu-2815	Big MOSFET Slide Switch with Reverse Voltage Protection, HP	2815Pololu	qty1	each$8.49	line$8.49	1 alternate · none drop-inCHANGES Pololu 2813 · $8.95 · 211the 2813 Big Pushbutton Power Switch HP: the same HP ratings, but its control is ON/OFF/CTRL pins, not the slide switch's ON pin: the arm chain's ARM output drives CTRL (check Pololu's pin logic before fitting)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/pendant_specs	Momentary switch, 35 mm, 6 ft cord, 3.5 mm mono plug	Specs IC (10000114)seller's no. Specs IC (10000114)AbleNetIn stock (2026-10-27)	qty1	each$75.00	line$75.00	1 alternate · 1 drop-inDROP-IN AbleNet Jelly Bean (10033400) · $75.00checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/pendant-hold-to-run	Specs IC switch - 1 3/8 in (3.5 cm) momentary button, 3.5 mm mono plug, 6 ft (183 cm) cord	Specs IC (SKU 10000114)seller's no. Product Number 10000114 (newsletter: 1000114)AbleNetschema.org InStock (structured data; no visible stock text) (2026-10-27)	qty1	each$75.00	line$75.00	3 alternates · 1 drop-inCHANGES Adafruit Product ID 423 (+ Product ID 2790 plug) · $10.00 · both 'In stock' (423: $7.50; 2790: $2.50)a foot switch (Adafruit 423, momentary, 250 VAC 10 A) wired to an Adafruit 2790 3.5 mm plug: $10 for bothchecked 2026-10-27DROP-IN AbleNet Jelly Bean (SKU 10033400, mpn 1.3.0001) · $75.00 · In stockchecked 2026-10-27CHANGES Coyote Direct Motion Raceworks 'Momentary Small Push Button w/6' Wound Cord' (no SKU on page) · $60.00	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							(Red or Green); $120.00 (Both) · OrderableEnds in two bare 18 AWG wires, so wire on a 3.5 mm mono plug (not included). Sealed 10 A momentary button with a 0.625 in body and a 6 ft coiled cord when pulled tight; lock nut and washer included, no bracket. A true hand-held pendant (transbrake-style button).checked 2026-10-27		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/contactor_lev100	Sealed DC contactor 100 A, 900 VDC, 12 V coil (5.5 W)	TE Kilovac LEV100A4ANG (3-1618389-7)seller's no. TE Kilovac LEV100A4ANG (3-1618389-7)TTI, Inc.301 In Stock (2026-10-27)	qty1	each$86.64	line$86.64	2 alternates · 2 drop-inDROP-IN TTI, Inc. TE EV200AAANA · $127.61 · 439 In Stockchecked 2026-10-27DROP-IN RS Americas LEV100A4ANG (RS 70062417) · $76.47checked 2026-10-27	☐	☐
☐	buy/resistor_10r_25w	10 ohm 25 W aluminium-housed resistor (box of 10)	TE HSA2510RJseller's no. TE HSA2510RJTTI, Inc.	qty1buy 1 pack of 10	each$3.70$37.00 per 10	line$37.00used $3.70	1 alternate · 1 drop-inDROP-IN Mouser 279-THS2510RJ (TE Connectivity THS2510RJ) · $7.19 · 208 in stock (Mouser's figure as listed on findchips.com today)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/resistor_120	120 ohm 1/4 W (1000)	YAGEO CFR-25JB-52-120Rseller's no. YAGEO CFR-25JB-52-120RTTI, Inc.	qty2buy 1 pack of 1000	each$0.0156$15.60 per 1000	line$15.60used $0.03	1 alternate · 1 drop-inDROP-IN Mouser 603-CFR-25JB-52-120R (YAGEO CFR-25JB-52-120R) · $0.10 · 19198 in stock (Mouser's figure as listed on findchips.com today)checked 2026-10-27	☐	☐
☐	buy/shunt_150a	Blue Sea meter shunt 150 A / 50 mV (manganin), in the battery negative	analog meter shunt 150 A / 50 mV (UPC 632085092313)go2marineavailable (2026-10-27)	qty1	each$48.74	line$48.74	1 alternate · 1 drop-inDROP-IN The Solar Store product page · $59.00 · available; Deltec MKB-100-100 (100 A / 100 mV)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/mrbf_80a	Blue Sea 5181 MRBF terminal fuse 80 A, 58 V DC, 2 kA at 58 V	5181go2marineavailable (2026-10-27)	qty2	each$20.50	line$41.00	1 alternate · 1 drop-inDROP-IN go2marine product page · $13.50 · available; Egis MRBF 80 A, 58 V DCchecked 2026-10-27	☐	☐
☐	buy/blue-sea-5191-mrbf-terminal-fuse-block	Blue Sea Systems 5191 MRBF Terminal Fuse Block 30-300 A	5191 MRBF Terminal Fuse Block 30-300Ago2marineavailable (2026-10-27)	qty1	each$31.50	line$31.50	1 alternate · 1 drop-inDROP-IN go2marine product page · $17.25 · available; Egis MRBF block, single output, 58 V DCchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/adafruit-feather-m4-can-4759	Feather M4 CAN Express with ATSAME51	4759Adafruit	qty1	each$24.95	line$24.95	2 alternates · 2 drop-inDROP-IN Digi-Key 1528-4759-ND · $24.95 · 473checked 2026-10-27DROP-IN Mouser 485-4759 · $24.95 · 455checked 2026-10-27	☐	☐
☐	buy/feather_m4_can	Feather M4 CAN Express (ATSAME51)	4759seller's no. 4759AdafruitIn stock (2026-10-27)	qty1	each$24.95	line$24.95	1 alternate · 1 drop-inDROP-IN Adafruit 5724 RP2040 CAN Bus Feather · $19.95checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/ina228	Adafruit INA228 85 V 20-bit power monitor breakout (removes its 15 mOhm shunt; wire it to the external shunt)	5832Adafruit58 in stock (2026-10-27)	qty1	each$14.95	line$14.95	1 alternate · 1 drop-inDROP-IN Adafruit product page · $10.95 · In stock; INA238 (16-bit), same boardchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/tof_vl53l1x	VL53L1X time-of-flight breakout, 30-4000 mm	3967seller's no. 3967AdafruitIn stock (2026-10-27)	qty1	each$14.95	line$14.95	1 alternate · 1 drop-inDROP-IN SparkFun SEN-14722 · $29.95checked 2026-10-27	☐	☐
☐	buy/can_pal	Adafruit CANPal (TJA1051T/3) CAN transceiver, switchable 120 Ohm	5708AdafruitIn stock (2026-10-27)	qty3	each$3.95	line$11.85	1 alternate · 1 drop-inDROP-IN Waveshare product page · $4.99 · InStock; SN65HVD230 boardchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/tool-microfit-crimper	Ratcheting crimper for open-barrel contacts, 18-28 AWG (Adafruit 1213): Micro-Fit 3.0 terminals (tool)	Product ID 1213seller's no. Product ID 1213Adafruit'In stock'; 'On Sale' $10.00 (struck-through $34.95; qty-break table still shows $34.95 for 1-9) (2026-10-27)	qty1	each$10.00	line$10.00	2 alternates · 2 drop-inDROP-IN Adafruit Product ID 350 · $49.95 · 'In stock'checked 2026-10-27DROP-IN Mouser 538-63819-0000 (Molex 63819-0000) · $501.22 · 40 in stock (Mouser); Heilind 42 in stock at $463.32checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/featherwing_proto	Feather Wing Proto	2884seller's no. 2884AdafruitIn Stock (2026-10-27)	qty2	each$4.95	line$9.90	1 alternate · 1 drop-inDROP-IN Mouser 485-2884 (Adafruit 2884) · $4.95 · 91 in stock (Mouser's figure as listed on findchips.com today)checked 2026-10-27	☐	☐
☐	buy/ferrules	Wire ferrules, 5000 pcs, AWG 22-7 (every screw terminal)	5000 pcs AWG 22-7Haisstronicaavailable (2026-10-27)	qty1	each$65.99	line$65.99	1 alternate · 1 drop-inDROP-IN Haisstronica product page · $33.99 · available (the tool)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/ferrule_crimper	Haisstronicaratchetingferrulecrimper, AWG 24-8	HX1574Haisstronicaavailable (2026-10-27)	qty1	each$33.99	line$33.99	1 alternate · 1 drop-inDROP-IN Haisstronica product page · $65.99 · available (the ferrule kit)checked 2026-10-27	☐	☐
☐	buy/wire_12awg	BNTECHGO 12 AWG silicone wire, red and black (VESC leads)	SW12G68008F10C2seller's no. SW12G68008F10C2BNTECHGO	qty2	each$14.98	line$29.96	2 alternates · 2 drop-inDROP-IN BNTECHGO SW12G68008F25C2 (25 + 25 ft) · $32.98checked 2026-10-27DROP-IN RaceDayQuads product page · $2.99 · 12 AWG sold out on 2026-10-27; 16/18/20 AWG in stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/wire_14awg	BNTECHGO 14 AWG silicone wire, red and black (RS-04 branches)	14 AWG silicone red + black 25 + 25 ftBNTECHGOnot shown (2026-10-27)	qty1	each$24.98	line$24.98	1 alternate · 1 drop-inDROP-IN RaceDayQuads product page · $2.99 · 14 AWG sold out on 2026-10-27checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/wire_8awg	BNTECHGO 8 AWG silicone wire, black, 10 ft (the pack leads, the MRBF, the shunt, the switch, the bus bars; red heat-shrink on every + end)	8 AWG silicone black 10 ftBNTECHGOnot shown (2026-10-27)	qty1	each$17.98	line$17.98	1 alternate · 1 drop-inDROP-IN BNTECHGO product page · $24.98 · not shown; 10 AWG 25 ft: carries the 80 A pulses, 55 A continuous (the MCU then caps the bus at 50 A)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/wire_22awg	BNTECHGO 22 AWG silicone wire, red and black (5 V, signals; twist the CAN pairs by hand)	22 AWG silicone red + black 25 + 25 ftBNTECHGOnot shown (2026-10-27)	qty1	each$11.48	line$11.48	1 alternate · 1 drop-inDROP-IN RaceDayQuads product page · $1.99 · 22 AWG sold out on 2026-10-27; 20 AWG in stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/wire_18awg	BNTECHGO 18 AWG silicone wire, red and black (converters, 24 V and 12 V)	18 AWG silicone red + black 10 + 10 ftBNTECHGOnot shown (2026-10-27)	qty1	each$9.48	line$9.48	1 alternate · 1 drop-inDROP-IN RaceDayQuads product page · $1.99 · 18 AWG in stockchecked 2026-10-27	☐	☐
☐	buy/proto-board-52x50	Double-sided prototype board 50 x 70 mm, cut to size	double-sided prototype board 50 x 70 mmseller's no. PRT-08619SparkFunIn stock (2026-10-27)	qty5	each$7.25	line$36.25	3 alternates · 3 drop-inDROP-IN Tayda Electronics product pagechecked ? DROP-IN Tayda Electronics product pagechecked ? DROP-IN Tayda Electronics product pagechecked ?	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/sparkfun-rob-13065	Slip Ring - 12 Wire (2A)	ROB-13065SparkFun	qty1	each$27.95	line$27.95	1 alternate · 1 drop-inDROP-IN Adafruit 1196 · $19.95 · 6checked 2026-10-27	☐	☐
☐	buy/hx711	HX711 load cell amplifier	SEN-13879seller's no. SEN-13879SparkFunIn stock (2026-10-27)	qty2	each$4.95	line$9.90	2 alternates · 2 drop-inDROP-IN Adafruit 5974 · $9.95checked 2026-10-27DROP-IN Adafruit 5974 · $9.95checked 2026-10-27	☐	☐
☐	buy/resistor_kit	Resistor kit 1/4 W, 500 (incl. 2.2k, 4.7k, 10k, 100k)	COM-10969seller's no. COM-10969SparkFun	qty1	each$9.00	line$9.00	1 alternate · 1 drop-inDROP-IN Mouser 474-COM-10969 (SparkFun COM-10969) · $9.95 · 112 in stock (Mouser's figure as listed on findchips.com today)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/proto-board-70x50	Double Side Prototyping Board 50x70mm	double-sided prototype board 50 x 70 mmseller's no. PRT-08619SparkFunIn stock (2026-10-27)	qty1	each$7.25	line$7.25	3 alternates · 1 drop-inCHANGES Botland Universal double sided board 50x70mm (Botland PLY-02745, EAN 5904422303303) · EUR 0.90 · OrderableSame board class as the primary. Mounting-hole positions are not given, so the stand's bosses may need to move (see notes).checked 2026-10-27CHANGES Adafruit Adafruit 866 'Snap-action 3-Wire Block Connector (12-24 AWG) - Pack of 3' / Adafruit 874 '5-Wire' (brand not confirmed as WAGO 221)No board: lever connectors splice the 24 V	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							feed (XT60 -> fuse -> switch -> DT12 plug power pins) and the pendant loop in free air. Drop the four M3 inserts and bosses for the 70 x 50 board (or keep them for a strain-relief plate), cable-tie the connectors to the floor or a printed cradle, and use 12-24 AWG wire. There is no screw-terminal board to service, and the pendant loop becomes two inline splices.checked 2026-10-27DROP-IN Tayda Electronics product pagechecked ?		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/diode_1n4001	Rectifier diode 1N4001 (1 A, 50 V)	COM-08589seller's no. COM-08589SparkFunIn stock (2026-10-27)	qty2	each$0.250	line$0.50	1 alternate · 1 drop-inDROP-IN Adafruit 755 (1N4001 Diode - 10 pack) · $1.50 / 10 · 38 in stockchecked 2026-10-27	☐	☐
☐	buy/bench_supply_36v	36 V 16.6 A enclosed supply	Mean Well LRS-600-36seller's no. Mean Well LRS-600-36Bravo Electro	qty1	each$61.60	line$61.60	1 alternate · 1 drop-inDROP-IN TRC Electronics LRS-600-36 · $66.80 · Contact Us (lead time from 1 week)checked 2026-10-27	☐	☐
☐	buy/dcdc_ddr30l12	DIN-rail DC-DC, 18-75 V in, 12 V 2.5 A out	Mean Well DDR-30L-12seller's no. Mean Well DDR-30L-12Bravo ElectroIn stock (2026-10-27)	qty1	each$27.60	line$27.60	1 alternate · 1 drop-inDROP-IN TRC Electronics DDR-30L-12 · $35.36 · 10 can ship nowchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/switchcraft-ta3flx	Mini-XLR 3-pin female cable plug (TA3FLX), latching, cable <= 4.32 mm	TA3FLXseller's no. 502-TA3FLXMouser323 (2026-10-27)	qty1	each$14.22	line$14.22	1 alternate · 1 drop-inDROP-IN Digi-Key TA3FLX-ND · $14.22 · 631checked 2026-10-27	☐	☐
☐	buy/switchcraft-tb3m	Mini-XLR 3-pin male panel receptacle (TB3M), latching, keyed	TB3Mseller's no. 502-TB3MMouser2619 (2026-10-27)	qty1	each$13.10	line$13.10	1 alternate · 1 drop-inDROP-IN Digi-Key TB3M-ND · $13.24 · 1488checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/littelfuse-0fhm0001zxj	In-line holder for a MINI blade fuse, black leads (MINI FHM series)	0FHM0001ZXJseller's no. 576-0FHM0001ZXJMouser8130 (2026-10-27)	qty1	each$9.47	line$9.47	1 alternate · 1 drop-inDROP-IN Digi-Key product pagechecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/te-dt04-12pa-l012	DT 12-way receptacle (pin side), key A, grey, flange mount (L012 mod) - LCSC C64235 85	DT04-12PA-L012seller's no. 571-DT04-12PA-L012Mouser7037 (2026-10-27)	qty1	each$7.38	line$7.38	5 alternates · 4 drop-inDROP-IN Newark DT04-12PA-L012 (Newark 38AH5620) · about $14.32 to $16.02 depending on quantity (search snippet); an oemstrade aggregator snippet showed Newark at $15.36 (1) / $14.91 (10) · In stock (99)checked 2026-10-27DROP-IN Del City DT04-12PA-L012 (Del City 15061229) · $12.84 at 1+ (search snippet) · In stockchecked 2026-10-27DROP-IN Chief Enterprises AT04-12PA-L012 (Amphenol Sine Systems) · $3.75 (250-	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							999), $3.46 (1000-9999); minimum purchase 250 units · Out of stock; 133 days lead timechecked 2026-10-27CHANGES Waytek AT04-12PA-PM11 (Amphenol Sine Systems; Waytek #38157)Different flange. Amphenol drawing AT04-12PX-PMXX shows a one-piece flange with gasket and 2 holes (2X dia 4.0, 16.4 deep) for #10 sheet-metal or ST 4.8 plastic screws (mounting plate dims 45.50 / 61.0 / 25.50 mm), not the 4-hole L012 flange. The turret-base cutout and screw holes would change.		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							It also needs the AW12P wedge.checked 2026-10-27DROP-IN LCSC C6423585 · $10.36checked 2026-10-27		
☐	buy/molex-43030-0010	Micro-Fit 3.0 female crimp terminal, tin, 26-30 AWG	43030-0010seller's no. 538-43030-0010Mouser287656 (2026-10-27)	qty30	each$0.200	line$6.00	1 alternate · 1 drop-inDROP-IN Digi-Key WM1838-NDchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/cap-mlcc-100n-50v	Capacitor 100 nF 50 V X7R radial ceramic, 2.5 mm leads	K104K15X7RF53L2seller's no. 594-K104K15X7RF53L2Mouser57498 (2026-10-27)	qty18	each$0.250	line$4.50	1 alternate · 1 drop-inDROP-IN Digi-Key BC3324-ND · $0.29checked 2026-10-27	☐	☐
☐	buy/tvs-p6ke15a	TVS diode 600 W unidirectional, 12.8 V standoff, 21.2 V clamp, DO-15 (P6KE15A)	P6KE15A-E3/54seller's no. 625-P6KE15A-E3Mouser2378 (2026-10-27)	qty3	each$0.960	line$2.88	1 alternate · 1 drop-inDROP-IN Digi-Key P6KE15A-E3/54GICT-ND · $0.96 · 1054checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/cap-mlcc-1u-50v	Capacitor 1 uF 50 V X7R radial ceramic, 5 mm leads	FG24X7R1H105KRT06seller's no. 810-FG24X7R1H105KRT6Mouser11666 (2026-10-27)	qty4	each$0.540	line$2.16	1 alternate · 1 drop-inDROP-IN Digi-Key 445-173374-1-ND · $0.51checked 2026-10-27	☐	☐
☐	buy/cap-film-100n-5pct	Capacitor 100 nF 63 V +-5 % PET film, 5 mm leads	MKS2C031001A00JSSDseller's no. 505-MKS2.1/63/5Mouser2166 (2026-10-27)	qty3	each$0.700	line$2.10	1 alternate · 1 drop-inDROP-IN Newark 46T6775 · $0.45 · 9338checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/schottky-1n5822	Schottky rectifier 3 A 40 V, DO-201AD (1N5822)	1N5822Gseller's no. 863-1N5822GMouser15695 (2026-10-27)	qty2	each$0.830	line$1.66	1 alternate · 1 drop-inDROP-IN Digi-Key 1N5822-E3/54GICT-ND · $0.91 · 7531checked 2026-10-27	☐	☐
☐	buy/yageo-mfr-25fbf52-100k	Resistor 100 k 1 % 1/4 W metal film (Yageo MFR-25FBF52)	MFR-25FBF52-100Kseller's no. 603-MFR-25FBF52-100KMouser15007 (2026-10-27)	qty16	each$0.100	line$1.60	1 alternate · 1 drop-inDROP-IN Digi-Key 100KXBK-NDchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/yageo-mfr-25fbf52-1k	Resistor 1.00 k 1 % 1/4 W metal film (Yageo MFR-25FBF52)	MFR-25FBF52-1Kseller's no. 603-MFR-25FBF52-1KMouser51348 (2026-10-27)	qty16	each$0.100	line$1.60	1 alternate · 1 drop-inDROP-IN Digi-Key 1.00KXBK-ND · $0.10checked 2026-10-27	☐	☐
☐	buy/yageo-mfr-25fbf52-10k	Resistor 10 k 1 % 1/4 W metal film (Yageo MFR-25FBF52)	MFR-25FBF52-10Kseller's no. 603-MFR-25FBF52-10KMouser114923 (2026-10-27)	qty14	each$0.100	line$1.40	1 alternate · 1 drop-inDROP-IN Digi-Key 10.0KXBK-ND · $0.10checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/ti-cd74hc423e	Dual retriggerable monostable, 74HC423 (no trigger on a rising reset), PDIP-16	CD74HC423Eseller's no. 595-CD74HC423EMouser453 (2026-10-27)	qty1	each$1.25	line$1.25	1 alternate · 1 drop-inDROP-IN Digi-Key 296-9229-5-ND · $1.22 · 296checked 2026-10-27	☐	☐
☐	buy/diode-1n5408g	Rectifier 3 A 1000 V, DO-201AD (1N5408)	1N5408Gseller's no. 863-1N5408GMouser33554 (2026-10-27)	qty2	each$0.580	line$1.16	1 alternate · 1 drop-inDROP-IN Newark 42K2941 · $0.62 · 31895checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/vishay-bat85	Schottky diode 30 V 200 mA, DO-35 (BAT85S)	BAT85S-TAPseller's no. 78-BAT85S-TAPMouser41995 (2026-10-27)	qty2	each$0.500	line$1.00	1 alternate · 1 drop-inDROP-IN Digi-Key BAT85S-TAPCT-ND · $0.50 · 61329checked 2026-10-27	☐	☐
☐	buy/ti-sn74hct14n	Hex Schmitt-trigger inverter SN74HCT14N (DIP-14)	SN74HCT14Nseller's no. 595-SN74HCT14NMouser3467 (2026-10-27)	qty1	each$0.920	line$0.92	1 alternate · 1 drop-inDROP-IN Digi-Key 296-8394-5-ND · $0.87checked 2026-10-27	☐	☐
☐	buy/schottky-1n5817	Schottky rectifier 1 A 20 V, DO-41 (1N5817)	1N5817-Tseller's no. 621-1N5817Mouser15381 (2026-10-27)	qty2	each$0.370	line$0.74	1 alternate · 1 drop-inDROP-IN Digi-Key 1N5817DICT-ND · $0.37 · 28734checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/omron-b3f-1000	Tactile switch 6x6 mm, 4.3 mm, 100 gf (B3F-1000)	B3F-1000seller's no. 653-B3F-1000Mouser25871 (2026-10-27)	qty2	each$0.350	line$0.70	1 alternate · 1 drop-inDROP-IN Digi-Key SW400-ND · $0.35checked 2026-10-27	☐	☐
☐	buy/bourns-mf-r010	Resettable PTC fuse 0.10 A hold, 60 V, radial (MF-R010)	MF-R010seller's no. 652-MFR010Mouser11219 (2026-10-27)	qty2	each$0.330	line$0.66	1 alternate · 1 drop-inDROP-IN Digi-Key MF-R010-ND · $0.40 · 7433checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/microchip-mcp120	Voltage supervisor 2.93 V, open-drain active-low reset, TO-92	MCP120-300DI/TOseller's no. 579-MCP120-300DI/TOMouser894 (2026-10-27)	qty1	each$0.600	line$0.60	1 alternate · 1 drop-inDROP-IN Digi-Key MCP120-300DI/TO-ND · $0.60 · 1901checked 2026-10-27	☐	☐
☐	buy/yageo-rsf100jb-73-2k2	Resistor 2.2k 1 W 5 % metal oxide (RSF100)	RSF100JB-73-2K2seller's no. 603-RSF100JB-73-2K2Mouser9082 (2026-10-27)	qty3	each$0.200	line$0.60	1 alternate · 1 drop-inDROP-IN Digi-Key 2.2KW-1-NDchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/yageo-mfr-25fbf52-39k	Resistor 39 k 1 % 1/4 W metal film (Yageo MFR-25FBF52)	MFR-25FBF52-39Kseller's no. 603-MFR-25FBF52-39KMouser5471 (2026-10-27)	qty5	each$0.100	line$0.50	1 alternate · 1 drop-inDROP-IN Digi-Key 39.0KXBK-NDchecked 2026-10-27	☐	☐
☐	buy/onsemi-1n4148	Diode 1N4148 (DO-35)	1N4148seller's no. 512-1N4148Mouser1090369 (2026-10-27)	qty4	each$0.100	line$0.40	1 alternate · 1 drop-inDROP-IN Digi-Key 1N4148FS-ND · $0.10checked 2026-10-27	☐	☐
☐	buy/yageo-mfr-25fbf52-100r	Resistor 100 R 1 % 1/4 W metal film (Yageo MFR-25FBF52)	MFR-25FBF52-100Rseller's no. 603-MFR-25FBF52-100RMouser49654 (2026-10-27)	qty4	each$0.100	line$0.40	1 alternate · 1 drop-inDROP-IN Digi-Key 100XBK-NDchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/yageo-mfr-25fbf52-6k8	Resistor 6.8 k 1 % 1/4 W metal film (Yageo MFR-25FBF52)	MFR-25FBF52-6K8seller's no. 603-MFR-25FBF52-6K8Mouser12179 (2026-10-27)	qty3	each$0.100	line$0.30	1 alternate · 1 drop-inDROP-IN Digi-Key 13-MFR-25FRF52-6K8CT-ND · $0.10checked 2026-10-27	☐	☐
☐	buy/yageo-mfr-25fbf52-4k7	Resistor 4.7 k 1 % 1/4 W metal film (Yageo MFR-25FBF52)	MFR-25FBF52-4K7seller's no. 603-MFR-25FBF52-4K7Mouser12937 (2026-10-27)	qty2	each$0.110	line$0.22	1 alternate · 1 drop-inDROP-IN Digi-Key product pagechecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/yageo-mfr-25fbf52-162r	Resistor 162 R 1 % 1/4 W metal film (Yageo MFR-25FBF52)	MFR-25FBF52-162Rseller's no. 603-MFR-25FBF52-162RMouser7377 (2026-10-27)	qty2	each$0.100	line$0.20	1 alternate · 1 drop-inDROP-IN Arrow V99:2348_17733694 · $0.02 · 8185 (Americas)checked 2026-10-27	☐	☐
☐	buy/yageo-mfr-25fbf52-182k	Resistor 182 k 1 % 1/4 W metal film (Yageo MFR-25FBF52)	MFR-25FBF52-182Kseller's no. 603-MFR-25FBF52-182KMouser31912 (2026-10-27)	qty2	each$0.100	line$0.20	1 alternate · 1 drop-inDROP-IN Quest Components product page · $0.05 · 440checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/yageo-mfr-25fbf52-1k33	Resistor 1.33 k 1 % 1/4 W metal film (Yageo MFR-25FBF52)	MFR-25FBF52-1K33seller's no. 603-MFR-25FBF52-1K33Mouser17999 (2026-10-27)	qty2	each$0.100	line$0.20	1 alternate · 1 drop-inDROP-IN Arrow V99:2348_17668792 · $0.02 · 3495 (Americas)checked 2026-10-27	☐	☐
☐	buy/yageo-mfr-25fbf52-20r	Resistor 20.0 R 1 % 1/4 W metal film (Yageo MFR-25FBF52)	MFR-25FBF52-20Rseller's no. 603-MFR-25FBF52-20RMouser3236 (2026-10-27)	qty2	each$0.100	line$0.20	1 alternate · 1 drop-inDROP-IN Arrow V99:2348_17665225 · $0.02 · 6692checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/yageo-mfr-25fbf52-324r	Resistor 324 R 1 % 1/4 W metal film (Yageo MFR-25FBF52)	MFR-25FBF52-324Rseller's no. 603-MFR-25FBF52-324RMouser7829 (2026-10-27)	qty2	each$0.100	line$0.20	1 alternate · 1 drop-inDROP-IN Arrow V99:2348_17662115 · $0.02 · 9514checked 2026-10-27	☐	☐
☐	buy/yageo-mfr-25fbf52-40r2	Resistor 40.2 R 1 % 1/4 W metal film (Yageo MFR-25FBF52)	MFR-25FBF52-40R2seller's no. 603-MFR-25FBF52-40R2Mouser5209 (2026-10-27)	qty2	each$0.100	line$0.20	1 alternate · 1 drop-inDROP-IN Quest Components product page · $0.06 · 557checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/yageo-mfr-25fbf52-60k4	Resistor 60.4k 1% 1/4W metal film (Yageo MFR-25FBF52)	MFR-25FBF52-60K4seller's no. 603-MFR-25FBF52-60K4Mouser12929 (2026-10-27)	qty2	each$0.100	line$0.20	1 alternate · 1 drop-inDROP-IN Digi-Key 60.4KXBK-NDchecked 2026-10-27	☐	☐
☐	buy/yageo-mfr-25fbf52-63k4	Resistor 63.4k 1% 1/4W metal film (Yageo MFR-25FBF52)	MFR-25FBF52-63K4seller's no. 603-MFR-25FBF52-63K4Mouser8863 (2026-10-27)	qty2	each$0.100	line$0.20	1 alternate · 1 drop-inDROP-IN Digi-Key product pagechecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/yageo-mfr-25fbf52-66k5	Resistor 66.5k 1% 1/4W metal film (Yageo MFR-25FBF52)	MFR-25FBF52-66K5seller's no. 603-MFR-25FBF52-66K5Mouser19592 (2026-10-27)	qty2	each$0.100	line$0.20	1 alternate · 1 drop-inDROP-IN Digi-Key 66.5KXBK-NDchecked 2026-10-27	☐	☐
☐	buy/yageo-mfr-25fbf52-69k8	Resistor 69.8k 1% 1/4W metal film (Yageo MFR-25FBF52)	MFR-25FBF52-69K8seller's no. 603-MFR-25FBF52-69K8Mouser14226 (2026-10-27)	qty2	each$0.100	line$0.20	1 alternate · 1 drop-inDROP-IN Digi-Key 13-MFR-25FBF52-69K8-NDchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/yageo-mfr-25fbf52-73k2	Resistor 73.2k 1% 1/4W metal film (Yageo MFR-25FBF52)	MFR-25FBF52-73K2seller's no. 603-MFR-25FBF52-73K2Mouser8644 (2026-10-27)	qty2	each$0.100	line$0.20	1 alternate · 1 drop-inDROP-IN Digi-Key 13-MFR-25FBF52-73K2-NDchecked 2026-10-27	☐	☐
☐	buy/yageo-mfr-25fbf52-80r6	Resistor 80.6R 1% 1/4W metal film (Yageo MFR-25FBF52)	MFR-25FBF52-80R6seller's no. 603-MFR-25FBF52-80R6Mouser18577 (2026-10-27)	qty2	each$0.100	line$0.20	1 alternate · 1 drop-inDROP-IN Digi-Key product pagechecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/bumper-feet-12.7	3M Bumpon SJ5003 - hemisphere, 0.440 in (11.2 mm) diameter x 0.200 in (5.1 mm) high	SJ5003 (Bumpon)seller's no. 517-SJ-5003BKMouser187488 (2026-10-27)	qty4buy 1 pack of 56	each$0.0023$0.13 per 56	line$0.13used $0.01	3 alternates · 1 drop-inCHANGES CabinetParts.com 3M Bumpon SJ5312 (clear cylinder, 12.7 x 3.6 mm)Correct 12.7 mm diameter (sits centred in the 13.2 mm recess) but only 3.6 mm tall, so it stands about 2.6 mm proud instead of 4.5. Update the foot envelope height in cad/turret_stand.py.checked 2026-10-27CHANGES Adafruit Adafruit 550 'Little Rubber Bumper Feet - Pack of 4' · $0.95 per pack of 4 (1-9 packs), $0.86 (10-99), $0.76 (100+) · In stockOnly about 8 mm diameter x 4 mm tall (style	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							varies by supplier). It would sit loose in the 13.2 mm recess and stand about 3 mm proud. Shrink the recesses to about 8.5 mm or accept off-centre feet.checked 2026-10-27DROP-IN Digi-Key product pagechecked ?		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/timer_offdelay	Timer relay, multifunction (off-delay with a trigger input), 0.1-10 s, 12-125 VDC / 240 VAC, 10 A	ProSense T2R-M1-ADJ-240Useller's no. ProSense T2R-M1-ADJ-240UAutomationDirect 76 Available (2026-10-27)	qty1	each$44.50	line$44.50	2 alternates · 2 drop-inDROP-IN DROK 200350 (LCD multifunction delay relay module, 6-30 V) · $6.55checked 2026-10-27DROP-IN Galco Finder 80.01.0.240.0000 · $78.85 · ships 11/6checked 2026-10-27	☐	☐
☐	buy/can_cable	Shielded 22 AWG 1.5-pair data cable, 20 ft	Quabbin Q0357-1seller's no. Quabbin Q0357-1AutomationDirect	qty1	each$21.20	line$21.20	1 alternate · 1 drop-inDROP-IN Pro Wire & Cable Belden 3106A · $82.60 · 823136 in stock; Typically Ships in 1-2 Dayschecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/main_switch	Flipsky Anti Spark Switch Smart Enhanced 300A V2.0, 12-84 V (3-20S), 100 A continuous, 16 mm button	ANSIFS20SHFlipsky ships from CNavailable (2026-10-27)	qty1	each$52.00	line$52.00	1 alternate · 1 drop-inDROP-IN Flipsky product page · $48.00 · available; Antispark Pro V3.0 (AWAPFS14SH), 12-60 V: 53.3 V full is inside itchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/load_cell_s500	S-type load cell 500 kg (C2), M12 x 1.75 both ends, 51 x 76 x 19 mm	3140_0seller's no. 3140_0PhidgetsQuantity Available: 358 (2026-10-27)	qty1	each$50.00	line$50.00	2 alternates · 2 drop-inDROP-IN RobotShop Phidgets S-Type Load Cell 500 kg (C3) · $71.54checked 2026-10-27DROP-IN SparkFun SEN-14282 (TAS501, 200 kg) · $96.95 · In stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/sunnysky-x2216-1400kv-v3	SunnySky X2216 1400KV V3 outrunner, short shaft, with its bolt-on prop adapter and X-mount	X2216 V3 1400KV, short shaft (SKU SS-XV3-X2216-S-1400-V3)seller's no. SS-XV3-X2216-S-1400-V3 (variant 27868073427038, '1400 Short')SunnySky USAavailable: true (Shopify; no count exposed) (2026-10-27)	qty2	each$24.99	line$49.98	2 alternates · 2 drop-inDROP-IN ReadyMadeRC SS-XV3-X2216-S-1400-V3 (RMRC 84546, GTIN 680306447800) · $24.99 · In Stock (no quantity shown)checked 2026-10-27DROP-IN ReadyMadeRC SS-XV3-X2216-S-1400-V3 (RMRC 84546, GTIN 680306447800) · $24.99 · In Stock (no quantity shown)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/dial_indicator	Dial indicator 0-1 in x 0.001 in with a magnetic base	G9849 (Grizzly Magnetic Base/Dial Indicator Combo)seller's no. G9849 (Grizzly Magnetic Base/Dial Indicator Combo)GrizzlyAvailable; Delivery est. by 11/2 - 11/5; Available in Bellingham, WA and Springfield, MO (2026-10-27)	qty1	each$49.95	line$49.95	1 alternate · 1 drop-inDROP-IN Shars 303-2724 · $66.95 · In stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/blue-sea-6006	m-Series Mini ON-OFF battery switch, single circuit, 48 V DC, 275-300 A continuous (D70: the stand's DC power switch, 15 A or more)	6006Walmart	qty1	each$34.16	line$34.16	2 alternates · 2 drop-inDROP-IN NavStore product page · $27.95checked 2026-10-28DROP-IN Waytek product page · $39.96checked 2026-10-28	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/dewalt-20v-dock-adapter-12awg	DeWalt 20 V MAX battery dock adapter, 12 AWG leads (takes a FLEXVOLT pack in its 20 V mode)	DeWalt 18/20 V MAX battery dock adapter, 12 AWG leadsWalmart	qty1	each$12.95	line$12.95	1 alternate · 1 drop-inDROP-IN Etsy product page · $13.50checked?	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/tool-deutsch-crimper-iwd1620	IWISS IWD-1620 crimper for Deutsch DT size-16 solid contacts, 16-20 AWG (tool)	IWD-1620 (Shopify variant 40373778415729)seller's no. IWD-1620 (Shopify variant 40373778415729)IWISS available: true (2026-10-27)	qty1	each$24.99	line$24.99	1 alternate · 1 drop-inDROP-IN McMaster-Carr 6084T3 · $64.98 · 'Delivers tomorrow 7-9 am'checked 2026-10-27	☐	☐
☐	buy/ferrule_kit	Ferrule crimper + 770 ferrules (23-10 AWG)	HSC8 6-4-KITseller's no. HSC8 6-4-KITIWISS	qty1	each$19.99	line$19.99	1 alternate · 1 drop-inDROP-IN Amazon (sold by iwiss, ships from Amazon) ASIN B07LCF39W9 · $24.99 · In Stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/worker-cheetah-rival-rounds-100	Worker Cheetah Rival-compatible rounds, 100-pack	Cheetah Rival-compatible rounds, 100-packOut of Darts	qty2	each$14.99	line$29.98	2 alternates · 2 drop-inDROP-IN Amazon ASIN B071SC9QYG · $14.99 · orderable (Add to cart shown); Amazon's Choice 'Overall Pick', 4.7 stars (8.9K), '400+ bought in past month'checked 2026-10-27DROP-IN Out of Darts product pagechecked ?	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/hasbro-nerf-rival-accu-round-60	Nerf Rival 60 Accu-Round Refill (Hasbro F5052), 60 official Rival rounds	Worker Cheetah 100-packOut of DartsIn stock (2026-10-27)	qty1	each$14.99	line$14.99	3 alternates · 3 drop-inDROP-IN Walmart (the catalogue's) F5052 · $19.99 eachchecked 2026-10-27DROP-IN Walmart Nerf Rival 50-Round Refill · $15.9550 rounds, not 60checked 2026-10-27DROP-IN eBay product page · $16.95checked ?	☐	☐
☐	buy/fuse_kldr030	Class CC time-delay fuse 30 A, 600 VAC / 300 VDC	Littelfuse KLDR030.TXPseller's no. Littelfuse KLDR030.TXPStandard Electric Supply812 In Stock (2026-10-27)	qty1	each$31.70	line$31.70	2 alternates · 2 drop-inDROP-IN Galco Littelfuse KLDR30 · $44.33 · 10 In Stockchecked 2026-10-27DROP-IN Galco Mersen ATDR30 · $19.96 · 31 In Stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/lug_8awg_m8	Seltermtinned copper lug 8 AWG, 5/16 in (M8) stud, 10 pcs (the MRBF block)	MD0856PSeltermavailable (2026-10-27)	qty4buy 1 pack of 10	each$1.61$16.15 per 10	line$16.15used $6.46	1 alternate · 1 drop-inDROP-IN go2marine product page · $4.95 / 3 · available; Ancor 12-10 AWG M8checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/lug_8awg_m6	Seltermtinned copper lug 8 AWG, 1/4 in (M6) stud, 10 pcs (crimped on 10 AWG with a sleeve)	MD0814PSeltermavailable (2026-10-27)	qty10buy 1 pack of 10	each$1.50$15.01 per 10	line$15.01	1 alternate · 1 drop-inDROP-IN go2marine product page · $4.95 / 3 · available; Ancor adhesive heat-shrink rings 12-10 AWG M6checked 2026-10-27	☐	☐
☐	buy/stud_m12x80	Fully threaded stud M12 x 75	91217237seller's no. 91217237FastenalLocal DC Inventory; Direct Ship: Ships TOMORROW (2026-10-27)	qty1	each$31.02	line$31.02	1 alternate · 1 drop-inDROP-IN BelMetric S12X150FCLZ · $6.88 · in stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/fuse_holder_mini_inline	Bussmann HHM inline MINI fuse holder, 2-30 A (rated 32 V DC: the 58 V fuse does the interrupting; label it "58 V fuses only")	HHMRockAutoAdd to Cart (2026-10-27)	qty13	each$2.28	line$29.64	2 alternates · 2 drop-inDROP-IN LCSC product page · $0.30 · 0 (Keystone 3570 PCB holder: the fuse-board upgrade)checked 2026-10-27DROP-IN LCSC product page · $0.30 · 0 (Keystone 3570 PCB holder: the fuse-board upgrade)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/fuse_holder_cc	Class CC fuse holder, 1-pole, DIN rail, 600 VAC/DC	Littelfuse LPSC0001Zseller's no. Littelfuse LPSC0001ZGalco170 In Stock (2026-10-27)	qty1	each$28.99	line$28.99	2 alternates · 2 drop-inDROP-IN Standard Electric Supply Littelfuse LFPSC0001Z · $26.48 · 62 In Stockchecked 2026-10-27DROP-IN Galco Mersen USCC1 · $23.18 · 68 In Stockchecked 2026-10-27	☐	☐
☐	buy/usb_can_u2c	USB to CAN adapter (candleLights_usb: SocketCAN)	BIGTREETECH U2C V2.1 (SKU 1030000107)seller's no. BIGTREETECH U2C V2.1 (SKU 1030000107)BIQU Equipment ships from CNorderable (2026-10-27)	qty1	each$23.99	line$23.99	2 alternates · 2 drop-inDROP-IN InnoMaker USB2CAN-Module (isolated) · $34.99checked 2026-10-27DROP-IN West3D FYSETC UCAN · $19.99checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/plywood_18	Sanded plywood 23/32 in, 2 x 4 ft project panel	Internet #202093836seller's no. Internet #202093836The Home Depot	qty1	each$23.76	line$23.76	1 alternate · 1 drop-inDROP-IN Lowe's Item #7710, Model #12611 · $27.50checked 2026-10-27	☐	☐
☐	buy/weight_plate_2p5	Cast iron standard plate, 2.5 lb, 1 in hole (129.5 x 20.3 mm)	RPB2-5seller's no. RPB2-5Body-Solid	qty2	each$6.65	line$13.30	1 alternate · 1 drop-inDROP-IN Amazon (CAP Barbell) ASIN B0FNPPH748 (4-pack) · $17.80 / 4checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/weight_plate_1p25	Cast iron standard plate, 1.25 lb, 1 in hole (101.6 x 10.2 mm)	RPB1-25seller's no. RPB1-25Body-Solid	qty2	each$3.33	line$6.66	1 alternate · 1 drop-inDROP-IN Amazon (CAP Barbell) ASIN B002BRAN66 · $18.97checked 2026-10-27	☐	☐
☐	buy/nucleo_g474re	ST Nucleo-64 NUCLEO-G474RE (the balance and power MCU: 3 FDCAN)	NUCLEO-G474RESTeStoreIn stock (1-2 per order) (2026-10-27)	qty1	each$19.90	line$19.90	1 alternate · 1 drop-inDROP-IN SparkFun product page · $31.50 · In stock; Teensy 4.1 (3 CAN) as the backup MCUchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/xt60h_f	Amass XT60H female, 5-pack (the robot's live ends: the charge port and the 48 V branch leads; D58)	XT60H-FPyrodroneavailable (2026-10-27)	qty6buy 2 packs of 5	each$0.798$3.99 per 5	line$7.98used $4.79	1 alternate · 1 drop-inDROP-IN RaceDayQuads product page · $0.99 · available (single)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/xt60h_m	Amass XT60H male, 5-pack (the devices' end: the VESC leads; D58)	XT60H-M 5-packPyrodroneavailable (2026-10-27)	qty2buy 1 pack of 5	each$0.798$3.99 per 5	line$3.99used $1.60	1 alternate · 1 drop-inDROP-IN RaceDayQuads product page · $0.99 · available (single)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/xt30_f	Amass XT30U female, 8-pack (the robot's live ends: the 24 V and 12 V branch leads; D58)	XT30U-F 8-packPyrodroneavailable (2026-10-27)	qty3buy 1 pack of 8	each$0.3738$2.99 per 8	line$2.99used $1.12	1 alternate · 1 drop-inDROP-IN RaceDayQuads product page · $0.99 · available (single)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/xt30_m	Amass XT30U male, 8-pack (the devices' end: the J501 and lidar box adapter leads; D58)	XT30U-M 8-packPyrodroneavailable (2026-10-27)	qty4buy 1 pack of 8	each$0.3738$2.99 per 8	line$2.99used $1.50	1 alternate · 1 drop-inDROP-IN RaceDayQuads product page · $0.99 · available (single)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/radio_stop_rx	RadioMaster ER3C-i ExpressLRS 2.4 GHz PWM receiver (the radio stop, D37: one PWM channel drives the RC relay switch)	ER3C-i 2.4 GHz ELRS PWMRadioMasteravailable (store JSON) (2026-10-27)	qty1	each$16.99	line$16.99	1 alternate · 1 drop-inDROP-IN RadioMaster product page · $18.99 · available; RP1 V2 nano, PWM on its UART pads (ELRS 3 setting)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/fuse_mini58_20a	Littelfuse MINI 58V blade fuse 20 A (the RS-04 branches)	0997020.WXNseller's no. C207028LCSC1,609 (2026-10-27)	qty9	each$0.610	line$5.49	2 alternates · 2 drop-inDROP-IN element14 product page · listed (snippet)checked 2026-10-27DROP-IN element14 product page · listed (snippet)checked 2026-10-27	☐	☐
☐	buy/fuse_mini58_10a	Littelfuse MINI 58V blade fuse 10 A (DDR-120C-24 input, the charge port)	0997010.WXNseller's no. C207026LCSC3,724 (2026-10-27)	qty6	each$0.410	line$2.46	2 alternates · 2 drop-inDROP-IN LCSC C207027 · $0.66 · 2,950 (15 A)checked 2026-10-27DROP-IN LCSC C207027 · $0.66 · 2,950 (15 A)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/fuse_mini58_7a5	Littelfuse MINI 58V blade fuse 7.5 A (the turret's feed W22 and the 12 V rail W43, D66)	099707.5WXNseller's no. C2680610LCSC ships from CN3,810 (2026-10-27)	qty4	each$0.5766	line$2.31	1 alternate · 1 drop-inDROP-IN LCSC (other suppliers) C2680610 · $0.31 · 3,000 (5-7 business days)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/fuse_mini58_30a	Littelfuse MINI 58V blade fuse 30 A (the VESC branches)	0997030.WXNseller's no. C207030LCSC ships from CN2,894 (2026-10-27)	qty4	each$0.440	line$1.76	1 alternate · 1 drop-inDROP-IN LCSC C207029 · $0.49 · 1,743 (25 A)checked 2026-10-27	☐	☐
☐	buy/gh_header_2p	JST GH 1.25 mm 2-pin header, top entry (SMT)	JST BM02B-GHS-TBT(LF)(SN) (LCSC C161690)seller's no. JST BM02B-GHS-TBT(LF)(SN) (LCSC C161690)LCSC ships from CN70,630 in stock (2026-10-27)	qty4	each$0.0897	line$0.36	1 alternate · 1 drop-inDROP-IN HQonline BM02B-GHS-TBT(LF)(SN) · $0.10checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/screw_m8x16_bh	M8 x 16 button head socket cap screw	13-8316seller's no. 13-831680/20 Inc.	qty16	each$0.610	line$9.76	1 alternate · 1 drop-inDROP-IN 80/20 Inc. 75-3422 · $1.14checked 2026-10-27	☐	☐
☐	buy/screw_m6x40	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M6 x 1 mm Thread Size, 40 mm Long	6544seller's no. 6544Bolt Depotno count shown; orderable (Add button, not marked 'Temporarily unavailable') (2026-10-27)	qty8	each$0.260	line$2.08	1 alternate · 1 drop-inDROP-IN Fastenal 1139571 · $21.50 / 50 · Local DC Inventory; Direct Ship: Ships TOMORROWchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/bolt_m12x80	Hex bolt M12 x 80, fully threaded (DIN 933)	6132seller's no. 6132Bolt Depotno count shown; orderable (Add button, not marked 'Temporarily unavailable') (2026-10-27)	qty1	each$1.99	line$1.99	1 alternate · 1 drop-inDROP-IN Fastenal 925206494 · $15.10 / 10 · FNL Supply Chain (not in the local DC); Direct Ship: Ships TOMORROWchecked 2026-10-27	☐	☐
☐	buy/screw_m12x30	Alloy steel socket head screw M12 x 30, class 12.9	6604seller's no. 6604Bolt Depotno count shown; orderable (Add button, not marked 'Temporarily unavailable') (2026-10-27)	qty2	each$0.960	line$1.92	1 alternate · 1 drop-inDROP-IN Fastenal 1139654 · $38.00 / 25 · Local DC Inventory; Direct Ship: Ships TOMORROWchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/nut_m6_nylock	Nylon-insert lock nut M6	4804seller's no. 4804Bolt Depotno count shown; orderable (Add button, not marked 'Temporarily unavailable') (2026-10-27)	qty8	each$0.100	line$0.80	1 alternate · 1 drop-inDROP-IN Fastenal 1140155 · $9.00 / 100 · Local DC Inventory; Direct Ship: Ships TOMORROWchecked 2026-10-27	☐	☐
☐	buy/screw_m8x40	Alloy steel socket head screw M8 x 40, class 12.9	6565seller's no. 6565Bolt Depotno count shown; orderable (Add button, not marked 'Temporarily unavailable') (2026-10-27)	qty2	each$0.380	line$0.76	1 alternate · 1 drop-inDROP-IN Fastenal 1139591 · $35.50 / 50 · Local DC Inventory; Direct Ship: Ships TOMORROWchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/nut_m12	Hex nut M12, DIN 934	4790seller's no. 4790Bolt Depotno count shown; orderable (Add button, not marked 'Temporarily unavailable') (2026-10-27)	qty2	each$0.240	line$0.48	1 alternate · 1 drop-inDROP-IN Fastenal 1140309 · $8.25 / 25 · Local DC Inventory; Direct Ship: Ships TOMORROWchecked 2026-10-27	☐	☐
☐	buy/washer_m12	Flat washer M12, DIN 125	4531seller's no. 4531Bolt Depotno count shown; orderable (Add button, not marked 'Temporarily unavailable') (2026-10-27)	qty2	each$0.080	line$0.16	1 alternate · 1 drop-inDROP-IN Fastenal 1140359 · $4.00 / 25 · Local DC Inventory; Direct Ship: Ships TOMORROWchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/usb-can-canable2	UCAN Board Based on STM32F072 USB to CAN Adapter Support with CAnable / CandleLight / Klipper firmware	UCAN (SKU MM-U2CAN-0001)seller's no. MM-U2CAN-0001 (variant 42187830067375)FYSETCavailable: true (Shopify; no count exposed) (2026-10-27)	qty1	each$8.00	line$8.00	5 alternates · 3 drop-inDROP-IN BIQU / BIGTREETECH 1030000107 (variant 39762747949154, '1 Pc / U2C V2.1 -3 CAN Output') · $23.99 · available: true (Shopify; no count exposed)checked 2026-10-27DROP-IN Amazon ASIN B0956NV6CM (UPC 781520762707) · $29.99 · Only 15 left in stock - order soonchecked 2026-10-27DROP-IN Openlight Labs CANable 2.0 (SKU PF-CANABLE-2.0) · $35.00 · Out of stockchecked 2026-10-27CHANGES BIQU Equipment BIGTREETECH U2C V2.1	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							(variant '1 Pc / U2C V2.1 -3 CAN Output', SKU 1030000107) · $23.99 · OrderableBoard size not checked against the stand's 50 x 20 x 12 mm envelope, and it has several CAN outputs ('-3 CAN Output'). The listing is shared with the EBB boards, so pick the U2C variant. Firmware is BTT's candleLight fork (gs_usb), so Linux gets can0 without slcand.checked 2026-10-27CHANGES Grid Connect PEAK-System PCAN-USB IPEH-002021 (Grid Connect GC-CAN-USB); opto-isolated IPEH-002022		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
							(GC-CAN-USB-ISO) · $256.00 (no isolation); $314.00 (with isolation) · OrderableAn enclosed dongle on a cable, not a board on the stand floor, so the back-wall USB-C slot and the 50 x 20 x 12 mm envelope do not apply. From general knowledge, not the page: it has a D-Sub 9 CAN connector, so it needs a DB9-to-wire lead, and it uses Linux's peak_usb SocketCAN driver instead of gs_usb or slcan. About 30 times the price of the UCAN.checked 2026-10-27		

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/jack_3p5	3.5 mm panel jack	Switchcraft 35PM2Aseller's no. Switchcraft 35PM2AMaster Electronics	qty1	each$6.95	line$6.95	2 alternates · 2 drop-inDROP-IN Markertek Calrad 30-493Achecked 2026-10-27DROP-IN Canford Canford 43-259 miniature jack socket, 3.5 mm 3-polechecked 2026-10-27	☐	☐
☐	buy/gh_lead_2p_300	JST GH 1.25 pre-crimped lead, socket to socket, 300 mm (CAN, UART)	AGHGH28K305TLC Electronics12,900 (2026-10-27)	qty8	each$0.520	line$4.16	1 alternate · 1 drop-inDROP-IN TLC Electronics product page · $0.48 · 6,000 (150 mm)checked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/amass-xt60e-m	XT60E-M panel-mount plug (male), 2 screw holes	Amass XT60E-Mseller's no. Amass XT60E-MMaplin ships from UK	qty2	each$1.50	line$3.00	2 alternates · 2 drop-inDROP-IN LCSC XT60E-M.G.Y (C3040679) · $0.61 · In stock (3,091)checked 2026-10-27DROP-IN PyroDrone F3-13155-BLK · $11.99 / 10 · in stockchecked 2026-10-27	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/fuse_mini58_5a	Littelfuse MINI 58V blade fuse 5 A (DDR-60L-12 input; the 24 V J501 and turret branches)	0997005.WXNFarnell5,995 in stock, 10-14 business days from US stock (search result, 2026-10-27) (2026-10-27)	qty2	each$0.980	line$1.96	1 alternate · 1 drop-inDROP-IN Dasenic product page · $0.39 · 17,902 in stock, $0.3878 at 25+ (search result, 2026-10-27)checked 2026-10-27	☐	☐
☐	material/pa6-cf	Bambu Lab PA6-CF	pa6-cfBambu Lab	qty6.07 kgbuy 7 kg	each$79.99 / kg	line$559.93used $485.57	Use the material specified by the print plan.	☐	☐
☐	material/petg	Bambu Lab PETG Basic	petgBambu Lab	qty3.19 kgbuy 4 kg	each$13.99 / kg	line$55.96used $44.57	Use the material specified by the print plan.	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	material/tpu95	Bambu Lab TPU 95A HF	tpu95Bambu Lab	qty0.45 kgbuy 1 kg	each$41.99 / kg	line$41.99used $18.79	Use the material specified by the print plan.	☐	☐
☐	material/tpu95-gyroid	Bambu Lab TPU 95A HF, printed as a gyroid crush lattice (about 15 % infill, 2 walls, 1 mm skins)	tpu95-gyroidBambu Lab	qty0.26 kgbuy 1 kg	each$41.99 / kg	line$41.99used $10.75	Use the material specified by the print plan.	☐	☐
☐	buy/?base_board_inserts	base_board_inserts	NO PART NUMBER	qty1	eachNO PRICE	line—	NO ALTERNATE	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Ordered	In hand
☐	buy/?base_board_screws	base_board_screws	NO PART NUMBER	qty1	eachNO PRICE	line—	NO ALTERNATE	☐	☐
☐	buy/?base_carrier_inserts	base_carrier_inserts	NO PART NUMBER	qty1	eachNO PRICE	line—	NO ALTERNATE	☐	☐
☐	buy/?base_carrier_screws	base_carrier_screws	NO PART NUMBER	qty1	eachNO PRICE	line—	NO ALTERNATE	☐	☐
☐	buy/?hopper_inserts	hopper_inserts	NO PART NUMBER	qty1	eachNO PRICE	line—	NO ALTERNATE	☐	☐
☐	buy/?laser_inserts	laser_inserts	NO PART NUMBER	qty1	eachNO PRICE	line—	NO ALTERNATE	☐	☐
☐	buy/?laser_screws	laser_screws	NO PART NUMBER	qty1	eachNO PRICE	line—	NO ALTERNATE	☐	☐